

The Bases of Legitimacy: How Leaders Are Chosen Shapes How They Should Persuade

(Authors' names have been blinded for peer review.)

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Abstract

We study how the legitimacy of leadership shapes cooperation in a repeated public-goods game in which leaders can influence behavior only through nonbinding recommendations and fixed-form messages. Our experiment varies two sources of legitimacy: how leaders are selected and how they persuade followers. Each source has a less and a more collective form. Leaders are selected by *random* assignment, by *designation* from a previous leader, or by *election* by group members. These procedures give group members an increasing role in choosing the leader. Leaders then justify their guidance either through *personal* appeals to their own judgment or through *communal* appeals to shared social norms. We find that the effectiveness of communal persuasion relative to personal persuasion declines as leader selection becomes more collective. Communal appeals produce higher cooperation than personal appeals under random selection, a smaller advantage under designated selection, and lower cooperation when leaders are elected. This interaction arises mainly from how followers respond to the leader's guidance. It does not arise from differences in who becomes leader or in how leaders carry out their role. The results suggest that collective bases of legitimacy can substitute for one another rather than reinforce each other. Leadership therefore depends on the fit between how leaders acquire authority and how they justify it.

JEL codes: C92, D91, H41, D02

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1 Introduction

Leadership shapes social and economic outcomes through both formal control and noncoercive influence. Some leaders possess the authority to assign responsibilities, alter incentive structures, and sanction noncompliance. Others lack these instruments of formal power but may still influence collective action by aligning expectations, offering advice, and setting an example. Such non-coercive leadership arises in a wide range of settings, from student groups and academic collaborations to professional communities and social movements. Because followers remain free to disregard their guidance, the effectiveness of such leadership depends on its legitimacy: whether followers regard a leader's guidance as appropriate and deserving of voluntary compliance (Tyler and Lind, 1992).

Legitimacy can arise from multiple bases (Beetham, 1991; Suchman, 1995; Johnson et al., 2006). Consider team leaders who must coordinate members through communication. A leader appointed without members' input may find that teammates discount her personal judgment but respond more favorably when she appeals to shared norms. By contrast, personal appeals from an elected leader may carry greater weight if members perceive her election as conferring a mandate from the team. This example highlights two aspects that are particularly relevant to how legitimacy is established: (i) how leaders come to occupy their positions, and (ii) how they persuade followers to accept their guidance. Importantly, the effects of these two aspects need not be independent.

Leader selection concerns the procedural basis of legitimacy: how authority is conferred.¹ Procedures that provide members with voice and participation confer authority on a more collective basis and may strengthen voluntary acceptance (Tyler, 2006; Scharpf, 1999). *Persuasion style* concerns the substantive basis of legitimacy: the grounds on which leaders justify their authority when guiding followers. Some leaders may appeal to their personal judgment, experience, or leadership role, whereas others may appeal to social norms and standards represented as shared by the group.² In our study, we examine how leader selection and persuasion style jointly shape leadership effectiveness in sustaining group cooperation.

Specifically, we focus on leader-selection procedures and persuasion styles that vary in their degree of collectiveness, a dimension particularly relevant to democratic institutions, where legit-

¹We use *authority* to refer simply to the role-based standing of a leader to issue guidance. This is distinct from sociological uses that link authority to legitimacy or distinguish it from coercive power (Weber, 1947; Blau, 1963).

²This distinction echoes Weber's contrast between legal-rational authority, grounded in impersonal rules and offices, and charismatic authority, grounded in the perceived qualities of a particular leader (Weber, 1947).

imacy often rests on collective participation and shared standards (Cohen, 1997).³ In our design, selection becomes more collective as group members gain a greater role in choosing the leader, while persuasion becomes more collective when leaders appeal to shared norms and standards rather than to their own judgment, experience, or role. This common dimension allows us to ask how collective bases of legitimacy provided by leader selection and persuasion interact. An additive view would predict that more collective selection and more collective persuasion each strengthen legitimacy, so that the two reinforce one another. Alternatively, the two may substitute: collective appeals may add little, or may even be counterproductive, when leader selection already rests on a strong collective basis. Our experiment distinguishes between these possibilities.

We test these ideas in a laboratory experiment using a repeated public-goods game in which leadership operates solely through communication. Leaders cannot reward, punish, or make binding commitments; they can only issue nonbinding contribution recommendations and send fixed-form messages. The design therefore isolates persuasion as the channel through which legitimacy affects followers' cooperative behavior.

The experiment uses a 3×2 between-subjects design. Along the first dimension, leaders are selected through increasingly collective procedures: random assignment (*Random*), designation by the previous leader (*Designated*), or election by group members (*Elected*).⁴ Along the second dimension, we vary persuasion style by restricting the messages leaders can use to influence group members. In the *Communal* condition, leaders appeal to shared social norms and collective standards; in the *Personal* condition, they appeal to their own judgment, experience, and leadership role.

Our results reveal a clear interaction between leader selection and persuasion style. The relative effectiveness of Communal persuasion declines as leader selection becomes more collective. While Communal persuasion generates higher cooperation than Personal persuasion under Random selection, this advantage narrows under Designated selection and reverses under Elected selection, where Personal persuasion is more effective. This pattern suggests that collective bases of legitimacy may substitute rather than simply reinforce one another.

To investigate the mechanisms, we first show that differences in leaders' observable charac-

³We use *collective* to describe the extent to which leadership draws on group participation or shared group standards, rather than external assignment or a leader's individual attributes and judgment. This usage draws on the broader individualism-collectivism literature, which contrasts orientations centered on group concerns with those centered on individual autonomy and personal attributes (Triandis, 1995; Oyserman et al., 2002).

⁴These leader selection procedures differ in the collective basis on which authority is conferred: *Random* involves no participant input, *Designated* transmits authority through a previous leader's ranking, and *Elected* gives all current group members a role in selecting the leader.

teristics across selection procedures do not account for the treatment effects. We then distinguish between two behavioral channels through which legitimacy may shape cooperation: how leaders exercise their authority and how followers respond to their guidance. We find little evidence that the interaction is driven by systematic differences in leaders’ recommendations or message choices; observed variation largely reflects responses to members’ past behavior. Instead, the interaction appears primarily in followers’ responses to guidance, affecting both their general willingness to cooperate and the weight they place on numerical recommendations. One interpretation is that selection and persuasion provide alternative bases for accepting guidance. When members have little role in selection, appeals to shared norms may provide a collective basis for influence. Election, by contrast, may itself confer a mandate to exercise personal judgment on the group’s behalf, making further appeals to shared norms less consequential.

These findings contribute to the experimental literature on collective action and the institutions that facilitate it.⁵ Across collective action settings, outcomes are shaped by intragroup communication (Balliet, 2010; Koch et al., 2021; Yeo and Zhuo, 2024), recommended play (Chaudhuri and Paichayontvijit, 2010), reward and punishment mechanisms (Balliet et al., 2011), and the broader interaction and institutional environment (Dal Bó et al., 2010; Peysakhovich and Rand, 2016). Most closely related to our paper is research on leadership as a mechanism for improving collective outcomes. In public-goods settings, leadership may mitigate free-riding through enforcement (Baldassarri and Grossman, 2011), leading by example (Potters et al., 2007; Gächter and Renner, 2018), or communication (Dannenberg, 2015; Levy et al., 2011). In coordination settings, leaders may guide groups toward efficient equilibria either by facilitating the effective use of private information (Brandts and Cooper, 2025) or by shaping expectations through communication and recommendations (Brandts and Cooper, 2007; Sahin et al., 2015).

Within this literature, two strands are especially relevant. One examines leader-selection procedures. Leaders selected through elections (Baldassarri and Grossman, 2011; Brandts et al., 2015) or appointed based on high ability (Chemin, 2021) have been found to improve collective outcomes. By contrast, the effects of self-selection are less consistent (Dannenberg, 2015; Arbak and Villeval, 2013). The other examines how leaders exercise influence through their actions and words. Leading by example has been found to be more effective than making nonbinding pledges (Dannenberg, 2015). Speeches containing more charismatic elements can encourage effort and cooperation (Antonakis et al., 2022) while empowering messages can outperform directive mes-

⁵For reviews of cooperation in social dilemmas, including public-goods and repeated prisoner’s-dilemma experiments, see Ledyard (1995); Chaudhuri (2011); Zelmer (2003); Dal Bó and Fréchette (2018). For reviews of experimental coordination games and coordination failure, see Devetag and Ortmann (2007); Cooper and Weber (2020).

sages under one-way, but not two-way, communication (Boulu-Reshef et al., 2020). Together, these studies show that leadership effectiveness depends on both how leaders attain authority and how they exercise influence, yet these dimensions are usually studied separately.

Our paper makes three contributions. First, we bring together two strands of literature that have largely developed separately: research on leader selection and research on how leaders persuade followers. Second, we extend the experimental literature on leader selection by considering three routes to leadership: random assignment, designation by a previous leader, and election. The inclusion of designation introduces a common but understudied form of leadership succession into this literature. Third, by organizing both leader selection and persuasion style along a common dimension of collectiveness, we show that the relative effectiveness of collective persuasion declines as leader selection becomes more collective.

More broadly, our results also speak to leadership in organizations, teams, and institutions where persuasion operates alongside formal enforcement. They suggest that legitimacy should not be understood solely as a property of a particular leader or institutional feature, but as arising from the broader configuration of procedures, practices, and expectations surrounding authority. Practices that appear independently legitimacy-enhancing – such as greater follower participation or stronger appeals to shared norms – need not reinforce one another. Institutional design and leadership practice should therefore consider these sources jointly rather than in isolation.

Section 2 describes our experimental design. Section 3 reports our results, and Section 4 concludes.

2 Experimental Design

The experiment consists of two main stages. In Stage 1, all participants complete a real-effort task, the outcomes of which provide information used in subsequent leader assignment. Importantly, the task incorporates exogenous variation in difficulty, introducing variation in observed performance that is independent of underlying ability. In Stage 2, participants take on fixed roles as candidates or members and play a repeated public goods game. For each member-group, one candidate is assigned as leader. Leaders do not make contributions and cannot reward or punish; they can only issue fixed-form recommendations and messages before members make their contribution decisions. This design allows us to study how effectively leaders influence cooperation through persuasion under different selection procedures and persuasion styles.

We employ a 3×2 between-subjects design that independently varies the leader selection and persuasion style experienced by members. Leader-selection procedures are randomized across

member-groups within sessions, while persuasion style is randomized across sessions. Along the first dimension, we vary the extent to which *leader selection* incorporates member participation, creating increasingly collective procedures ranging from random assignment (Random), designation by a previous leader (Designated), to direct election by group members (Elected). Along the second dimension, we vary *persuasion style* by restricting the messages leaders may use to influence members. The two styles differ in whether leaders appeal primarily to their own judgment, experience, and leadership role (Personal) or to shared social norms and collective standards (Communal). Figure 1 summarizes the treatment conditions and experimental structure. Appendix D provides the complete experimental materials.

2.1 Stage 1: Addition Task

In Stage 1, participants complete a two-minute real-effort task consisting of a series of addition problems. Each problem requires them to sum two numbers ranging from 0 to 99, and each correct answer earns 0.5 tokens. The resulting scores are later displayed to members as performance information about potential leader candidates during leader assignment. Importantly, participants are not informed of this subsequent use when completing the task, preventing their effort from responding to anticipated roles or selection procedures.⁶

To introduce controlled variation in task difficulty, participants are randomly assigned to one of two task versions. In the easy version, both numbers to be summed are single-digit with probability 50%; in the hard version, this probability is 35%. Incorrect answers do not affect the difficulty of subsequent problems, preventing participants from strategically avoiding difficult questions.⁷ This variation is not disclosed to participants. It mimics the luck in task difficulty that would naturally arise if all participants drew problems from a common distribution, while giving the experimenter control over its size.

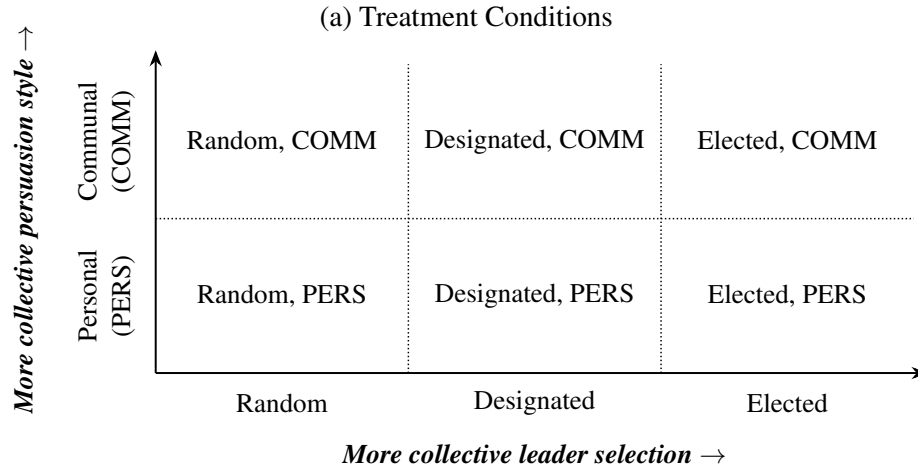
This design serves two purposes. First, it introduces random variation in task performance that is independent of underlying ability, thereby limiting systematic sorting on leader quality. Second, it increases the dispersion of observable scores, making performance information more salient during leader assignment.⁸

⁶Consistent with this, task scores do not differ significantly across the six treatment conditions (Kruskal–Wallis test, $p = 0.683$).

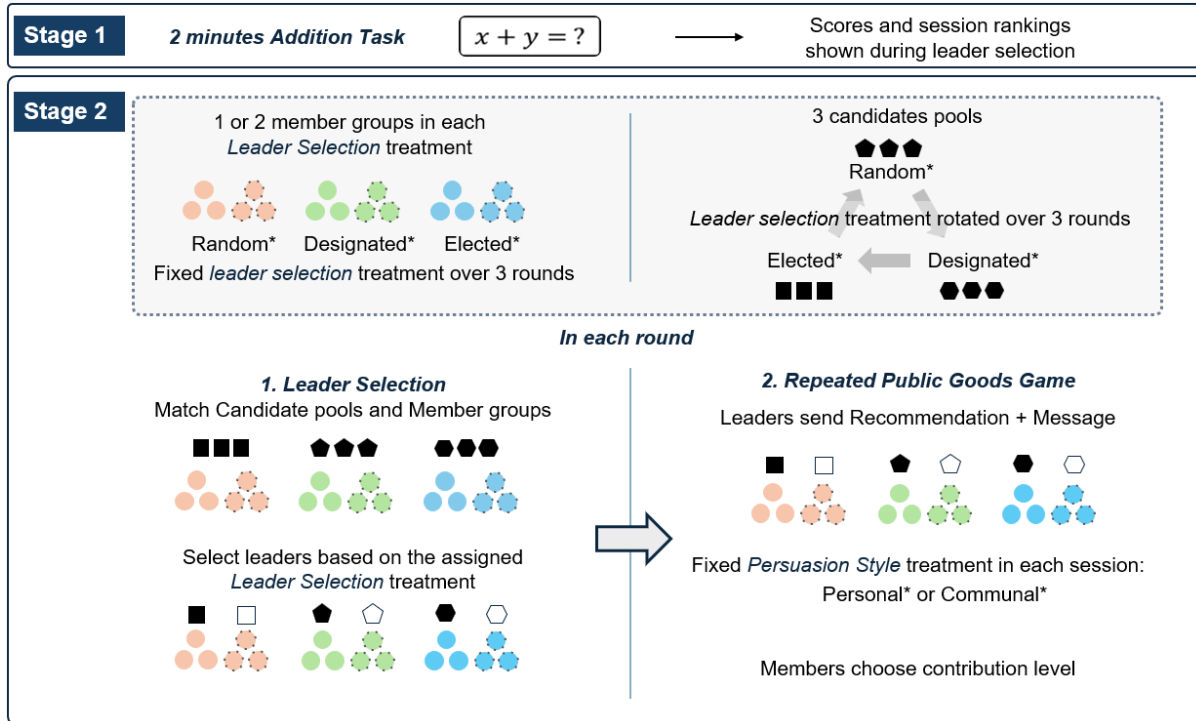
⁷The difficulty category of the next problem (single-digit sum or not) is redrawn only after a correct answer; following an incorrect answer, the next problem is drawn from the same category.

⁸Participants assigned to the easy version score higher on average than those assigned to the hard version (39.11 versus 33.60, $p < 0.001$). The distributions nevertheless overlap substantially (Figure A.1), making assigned difficulty hard to infer from a participant’s score alone. Table A.1 also shows no score differences between candidates

Figure 1: Treatment Conditions and Experimental Structure



(b) Experimental Flow for Each Session



2.2 Stage 2: Repeated Public-Goods Game

Stage 2 consists of a repeated public-goods game with two fixed roles: *candidates* and *members*. Participants are randomly assigned to one role at the beginning of Stage 2 and retain it throughout the session. Members are placed in groups of three (*member-groups*), while candidates are placed in pools of three (*candidate-pools*). Each member-group is matched with a candidate-pool from which its leader is selected.

The game comprises three rounds of ten periods each. Member-group composition remains fixed across all three rounds, while each group is matched with a different candidate-pool in each round. A new leader is therefore selected at the beginning of every round.

Leader Selection

At the beginning of each round, each candidate-pool is matched at random with one or two member-groups (depending on session size) that share the same leader-selection procedure. For each member-group, one candidate is then selected as leader under one of three procedures:

Random: Leaders are selected randomly by the computer. This is the least collective baseline procedure because authority is conferred exogenously, without input from members or previous leaders.

Designated: Leaders are selected according to the ranking submitted by the group's leader from the previous round. This procedure is intermediate in collectiveness because selection is made by a prior leader, but not by group members. In the first round, when no previous leader exists, the experimenter selects the leader at random; participants are informed that the assignment is based on the experimenter's preferences.

Elected: Leaders are selected based on members' rankings. This is the most collective procedure because all group members participate directly in leader selection. Rankings are converted into points (three for first, two for second, one for third), summed within each group, and the highest-scoring candidate is assigned as leader, with ties broken at random.

Table 1 reports the exact descriptions of the assignment procedures shown to members.

and members.

Table 1: Descriptions of Leader Assignment Shown to Members

	Round	Your group will now be assigned a leader from the new pool of candidates...
Random	1, 2, 3	<i>The leader will be assigned based on a random draw by the computer.</i>
Designated	1	<i>The leader will be assigned based on the experimenter's preferences.</i>
	2, 3	<i>The leader will be assigned based on your past leader, X's preferences.</i>
Elected	1, 2, 3	<i>The leader will be assigned based on your group's preferences.</i>

In all three conditions, members observe the candidates' addition task scores and their relative score rankings (top/middle/bottom) within the session before submitting a ranking of the candidates. Members submit rankings under every procedure, but these rankings determine leader selection only in the Elected condition. Individual rankings are private and are not revealed to other members. For member-groups assigned to the Designated condition, the group's previous leader observes similar information and ranks the current candidate-pool; this ranking determines the group's next leader.

When two member-groups are matched to the same candidate-pool, the relevant rankings may identify the same top-ranked candidate in the Designated and Elected conditions. In such cases, one group is selected at random to receive its second-ranked candidate.⁹ This procedure implements the submitted rankings as closely as possible while allowing two groups to share a candidate-pool.

Members' Decisions

Members participate in a standard voluntary contribution mechanism. In each period, members receive an endowment of 20 tokens and choose an integer contribution $0 \leq C_i^M \leq 20$ to invest in a joint project. Member i 's payoff is:

$$\pi_i^M = 20 - C_i^M + 0.5 \sum_{j=1}^3 C_j^M.$$

Before making contribution decisions in each period, members observe a message sent by their assigned leader. They also report their belief about the average contribution of the other members in their group and receive a 0.5-token bonus if their belief is within 2 tokens of the realized average.

⁹Because rankings are private, members cannot determine whether this constraint affected their assignment.

Leaders' Decisions

Candidates selected as leaders do not make contribution decisions, so that their influence operates exclusively through guidance rather than through leading by example. In each odd-numbered period, leaders send their member-group a fixed-form communication that also remains applicable in the following even-numbered period. Each communication contains two elements: (i) a numerical contribution recommendation and (ii) a substantive statement selected from a fixed set of messages. The available statements depend on the persuasion-style condition and differ in the grounds used to justify the leader's guidance:

Personal (PERS): Messages appeal to the leader's own judgment, experience, and role, representing a less collective persuasion style.

Communal (COMM): Messages appeal to shared social norms and collective standards, representing a more collective persuasion style.

Communication occurs once every two periods to avoid excessive repetition and limit the time devoted to it. The fixed message set also makes communication content comparable across leaders and limits variation arising from leaders' own wording or communication ability.

Table 2 presents the complete message sets, whose general structure is adapted from [Bouloureshef et al. \(2020\)](#). For later analysis, we also classify additional messages 1–8 into three categories: *Encouraging*, *Corrective*, and *Punitive*. Encouraging messages use positive reinforcement or motivational language; corrective messages call for adjustment toward improved outcomes in a relatively neutral manner; and punitive messages rely on blame, pressure, or exclusion. Following the GABRIEL framework proposed by [Asirvatham et al. \(2026\)](#), we use GPT as a standardized coding tool to apply these prespecified categories consistently across messages. Appendix C provides further details.

Leaders receive a fixed payment of 15 tokens per period plus a share of the member-group's public good returns:

$$\pi_i^L = 15 + 0.2 \sum_{j=1}^3 C_j^M.$$

The lower marginal return was chosen to limit perceptions that their recommendations are driven by personal gain while preserving incentives for group efficiency. In every period, leaders also report their belief about the average contribution of members in their group and receive a 0.5 token bonus if their estimate is within 2 tokens of the realized average.

Table 2: Messages by Persuasion-Style Condition

Category	Description of messages	Communal	Personal
<i>Compulsory Message</i>			
-	A message to set the tone and expectations	It is my role to tell you how you can draw from <i>norms in society</i> in order to achieve high levels of investments and earnings.	It is my role to tell you how you can draw from <i>my experience</i> in order to achieve high levels of investments and earnings.
-	Recommendation	Speaking as <i>a similar member of society</i> , I suggest that each of you contribute the following amount of tokens: []	Speaking as <i>leader of your group</i> , I suggest that each of you contribute the following amount of tokens: []
<i>Additional Message</i>			
Encouraging	After investments have increased	1. Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that <i>society has laid out for us</i> .	1. Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that <i>I have laid out</i> .
Corrective	After investments have decreased	2. We need to reverse this downward trend. By following <i>norms in society</i> , we can achieve a better outcome if we all invest more.	2. We need to reverse this downward trend. By following <i>my strategy</i> , we can achieve a better outcome if we all invest more.
Corrective	After some increased and some decreased their investments	3. We are getting there, but if <i>we think more about our societal standards</i> and invest more, we will all benefit.	3. We are getting there, but if <i>you follow my guidelines</i> and invest more, we will all benefit.
Encouraging	After high target investment levels have been reached	4. We have achieved <i>a goal worthy of members of society</i> which we must sustain.	4. We have achieved <i>my goal of high investments and high earnings</i> , which we must sustain.
Corrective	After investments have become stuck at a minimal, disappointing level	5. This situation <i>is not befitting the norms of society</i> . We should change course and invest more.	5. This situation <i>is not following my aims for the group</i> . We should change course and invest more.
Punitive	Desire to punish	6. You should be ashamed of <i>your actions which violate norms of society</i> .	6. You should be ashamed of <i>not following my instructions</i> .
Encouraging	Desire to reward	7. You have done well to <i>uphold the norms of society</i> .	7. You have done well to <i>follow my instructions</i> .
Punitive	Desire to exclude	8. We are trying to maximise our investments and earnings. Unless you <i>abide by the norms of society</i> , you are not part of the group.	8. We are trying to maximise our investments and earnings. Unless you <i>follow my instructions</i> , you are not part of the group.
-	-	9. No message	9. No message

Unassigned Candidates

Candidates not selected as leaders in a given round act as *shadow leaders*. Shadow leaders make the same recommendation and message decisions as selected leaders, but their choices are hypothetical and are not transmitted to members. Each shadow leader is matched with a selected leader from the same candidate-pool and observes the same information.

Shadow leaders receive a fixed payment of 10 tokens plus the same share of the member-group’s public good returns:

$$\pi_i^{SL} = 10 + 0.2 \sum_{j=1}^3 C_j^M.$$

They also report beliefs about members’ average contributions under the same incentive scheme as selected leaders. This procedure ensures that all candidates accumulate comparable experience over the course of the experiment.

2.3 Experimental Procedures

A total of 417 participants were recruited via email at a large university in Singapore (blinded for review) and took part in 16 sessions conducted in October 2022. The experiment was programmed in z-Tree (Fischbacher, 2007) and received ethical approval from the university. Each session consisted of exactly three candidate-pools and between four and six member-groups, resulting in session sizes of 21 to 27 participants (mean = 26). Table 3 reports treatment assignments and sample sizes.

Within each session, participants were randomly assigned to either the easy or hard version of the Stage 1 addition task. They were subsequently assigned at random to the role of member or candidate and placed in a fixed member-group or candidate-pool.

Table 3: Treatment Assignments and Sample Sizes

	Random	Designated	Elected
Communal persuasion	$n_{member} = 48$ $n_{candidate} = 72$; candidates experience all three selection procedures	$n_{member} = 45$	$n_{member} = 45$
Personal persuasion	$n_{member} = 42$ $n_{candidate} = 72$; candidates experience all three selection procedures	$n_{member} = 45$	$n_{member} = 48$

Member-groups are assigned to one of the six combinations of leader-selection procedure and persuasion style. Candidate-pools are assigned to one persuasion-style condition, but experience all three leader-selection procedures when rematched with member-groups across rounds.

Leader-selection procedures (*Random*, *Designated*, and *Elected*) were randomized across member-groups within each session, with every session containing all three procedures. Each member-group retained the same leader-selection procedure throughout Stage 2. Persuasion style (*Communal* or *Personal*) was randomized across sessions, so that all member-groups and candidate-pools within a session faced the same persuasion-style condition. Each member-group therefore experienced a fixed combination of leader-selection procedure and persuasion style throughout Stage 2. Candidate-pools were matched with different member-groups across rounds and hence experienced each of the three leader-selection procedures once, in randomized order.¹⁰

Upon arrival, participants were seated at computer terminals at random. The experimenters distributed and read aloud the instructions for Stage 1, after which participants completed the addition task. The instructions for Stage 2 were then distributed and read aloud. Before proceeding, participants were required to answer a set of comprehension questions correctly and complete a practice round involving simulated group members. They then played an anonymous, one-shot public-goods game with the same parameters as the Stage 2 game, providing a measure of baseline cooperativeness. Participants subsequently completed the three rounds of the repeated public-goods game, each comprising ten periods. At the end of each round, members completed a short questionnaire eliciting their perceptions of the leader assigned to their group before proceeding to the next round.

After Stage 2, participants completed a post-experimental survey eliciting demographic characteristics, attitudes toward authority, cooperation in groups, locus of control, competitiveness, and perceptions of legitimacy. The survey also included open-ended questions about participants' decision-making. At the end of the session, participants received a summary of their earnings. Payment was based on their Stage 1 earnings and three randomly selected periods from each round of Stage 2, converted at a rate of 20 tokens to SGD 1. Including a SGD 3 show-up fee, participants earned an average of SGD 15.35 for approximately 90 minutes of participation.

¹⁰Because leader-selection procedure remains fixed for member-groups, the rotation of candidate-pools does not alter the treatment comparisons for our main outcome, members' contributions. We also find no systematic differences in leaders' behavior across selection procedures and no evidence of order effects.

3 Results

3.1 Analysis Approach

Our primary outcome is members' contributions to the public good, normalized to lie between 0 and 1. We observe 8,190 member-period observations across 91 member-groups. For member i in group g , round r , and period p within the round, our main specification is

$$\begin{aligned} C_{igrp} = & \alpha + \beta_1 \mathbf{1}\{\text{Designated}_g\} + \beta_2 \mathbf{1}\{\text{Elected}_g\} \\ & + \gamma_1 \mathbf{1}\{\text{Random}_g\} \mathbf{1}\{\text{Communal}_g\} + \gamma_2 \mathbf{1}\{\text{Designated}_g\} \mathbf{1}\{\text{Communal}_g\} \\ & + \gamma_3 \mathbf{1}\{\text{Elected}_g\} \mathbf{1}\{\text{Communal}_g\} + \rho_r + \tau_p + \varepsilon_{igrp}. \end{aligned}$$

where $\mathbf{1}\{\cdot\}$ denotes an indicator for the corresponding treatment condition, and ρ_r and τ_p denote round and period-within-round fixed effects, respectively. The coefficients β_1 and β_2 capture under Personal persuasion, the differences between Designated and Random selection, and between Elected and Random selection, respectively. The coefficients γ_1 – γ_3 capture the effect of Communal relative to Personal persuasion within the Random, Designated, and Elected selection procedures, respectively.

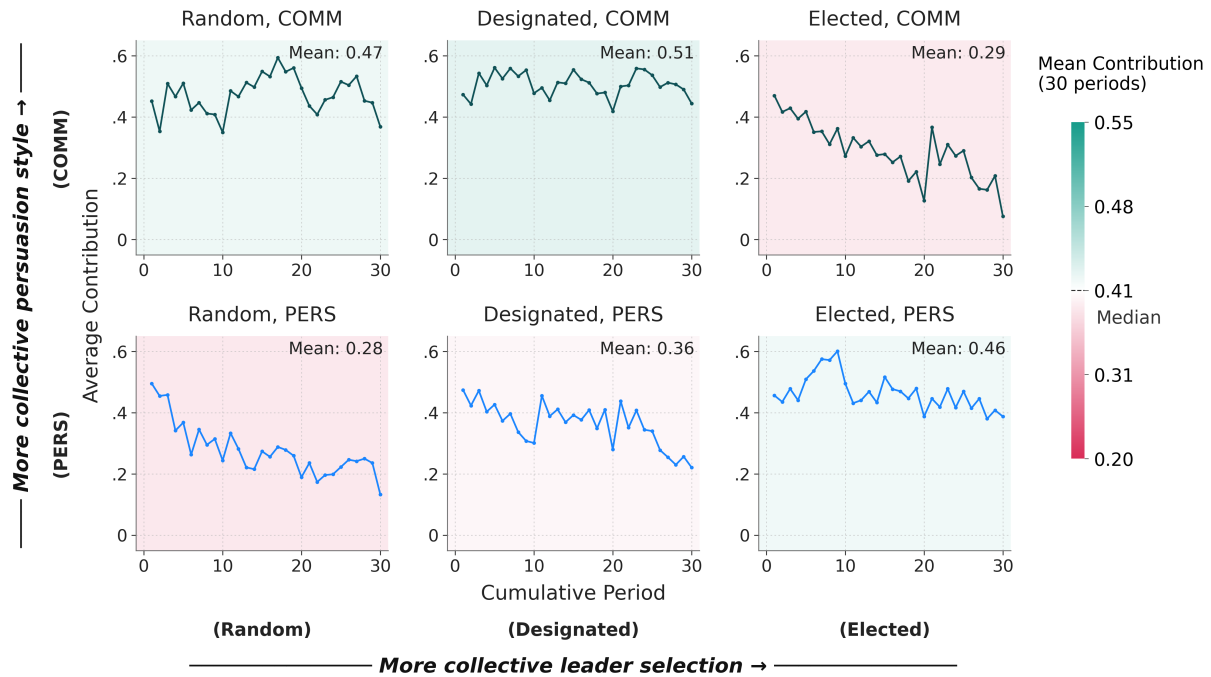
Organizing the analysis around these conditional effects speaks directly to whether the effectiveness of persuasion style depends on how authority is conferred. Differences among coefficients γ_1 – γ_3 formally test whether the effect of persuasion style varies across leader-selection procedures. We also analyze leaders' contribution recommendations and message choices in the odd-numbered periods where these decisions are made. Recommendations are likewise normalized to lie between 0 and 1. Optional messages 1–8 are classified into the three categories described in Table 2: *Encouraging*, *Corrective*, and *Punitive*. This yields 1,365 leader-period observations across 48 candidate-pools.

To account for the session-level assignment of persuasion style and potential dependence within sessions, all regression analyzes report standard errors clustered at the session level. As an additional robustness check, Appendix B also replicates Tables 4, 5, A.5, A.6, A.7, and A.8, with statistical significance assessed using wild-cluster bootstrap p -values.

3.2 Treatment Effects

We first examine how cooperation varies across the six combinations of leader selection and persuasion style, focusing on whether the effect of persuasion style depends on the leader-selection procedure. Figure 2 plots mean contributions over the 30 periods for each treatment condition. Contributions in the first period are statistically indistinguishable across the six conditions (joint test, $p = 0.977$), indicating comparable initial contribution levels.¹¹ Over time, however, the contribution paths diverge substantially, and by the final period, contributions differ significantly across the six conditions (joint test, $p = 0.044$).

Figure 2: Members' Contributions in Each Period



Notes: Each panel reports mean member contributions by cumulative period for one treatment condition. Contributions are normalized to $[0, 1]$ by dividing them by the 20-token endowment. Panel backgrounds are shaded according to the treatment condition's mean contribution across all 30 periods. See also Figure A.2 for the same plots split by leader selection procedure.

The divergence reveals a clear interaction between leader selection and persuasion style. As leader selection becomes more collective, the relative effectiveness of Communal persuasion declines. Under Random selection, the least collective procedure, Communal persuasion gen-

¹¹Baseline cooperativeness, measured by contributions in an initial anonymous one-shot public-goods game, also does not differ across treatment conditions (joint test, $p = 0.552$). The results remain qualitatively unchanged when controlling for baseline cooperativeness.

erates higher contributions than Personal persuasion. This advantage narrows under Designated selection and reverses under Elected selection, where Personal persuasion generates higher contributions. Figure 2 also illustrates the complementary interpretation of the interaction. Under Personal persuasion, contributions tend to increase as leader selection becomes more collective, whereas under Communal persuasion, contributions are lowest when leaders are elected. Thus, sources of legitimacy with similarly collective bases do not simply reinforce one another; instead, the pattern is consistent with substitution between the collective bases of selection and persuasion.

In the analysis that follows, we organize comparisons around the effect of persuasion style within each leader-selection procedure and then compare these conditional effects across procedures to formally test the interaction.

Table 4 reports the corresponding OLS estimates. Column 1 shows no aggregate effects of selection procedures or persuasion style in isolation. Column 2 allows the effect of persuasion style to vary across selection procedures and confirms the pattern in Figure 2. Relative to Personal persuasion, Communal persuasion increases contributions by 19.5 percentage points under Random selection ($p = 0.067$) and by 14.5 percentage points under Designated selection ($p = 0.118$), although the latter estimate is not statistically significant at conventional levels. Under Elected selection, by contrast, Communal persuasion reduces contributions by 17.6 percentage points ($p = 0.036$).¹²

Comparisons of these conditional effects provide formal evidence of the interaction. The effect of Communal relative to Personal persuasion differs significantly between Elected and Random selection ($p = 0.018$) and between Elected and Designated selection ($p = 0.021$), while the corresponding difference between Random and Designated selection is statistically insignificant. Column 3 adds the round and period-within-round fixed effects in our preferred specification; the estimates are unchanged.

Result 1. *The effect of persuasion style on cooperation depends on how the leader is selected. Relative to Personal persuasion, Communal persuasion increases contributions under Random selection, has a smaller and statistically insignificant effect under Designated selection, and significantly reduces contributions under Elected selection.*

¹²Appendix Figure A.3 and Table A.2 show similar patterns for members' beliefs about other members' contributions. The interaction between leader selection and persuasion style observed for contributions also appears in contribution beliefs, suggesting that treatment effects are accompanied by corresponding shifts in expectations about others' cooperation.

Table 4: Cooperation across Treatments

<i>Dep: Contribution Level</i>	(1)	(2)	(3)
Designated	0.056 (0.056)	0.086 (0.069)	0.086 (0.069)
Elected	0.000 (0.085)	0.187* (0.104)	0.187* (0.104)
COMM	0.052 (0.050)		
Random $\times$ COMM		0.195* (0.099)	0.195* (0.099)
Designated $\times$ COMM		0.145 (0.087)	0.145 (0.087)
Elected $\times$ COMM		-0.176** (0.076)	-0.176** (0.076)
Constant	0.353*** (0.067)	0.277*** (0.083)	0.277*** (0.083)
Round, Period-within-Round Fixed Effects			✓
<i>Differences in Persuasion Effects</i>			
(Random $\times$ COMM – Designated $\times$ COMM)		0.050 (0.110)	0.050 (0.110)
(Designated $\times$ COMM – Elected $\times$ COMM)		0.320** (0.124)	0.320** (0.124)
(Random $\times$ COMM – Elected $\times$ COMM)		0.370** (0.139)	0.370** (0.139)
Observations	8190	8190	8190

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 1 to Period 10 dummies. Contribution levels are normalized to $[0,1]$ by dividing contributions by the 20-token endowment.

3.3 Mechanisms

We next examine the mechanisms underlying the interaction between leader selection and persuasion style. We first ask whether the interaction can be explained by differences in the observable characteristics of leaders selected under different procedures. We then examine two behavioral channels: whether the treatments change how leaders exercise their authority and how members respond to leaders' guidance.

Characteristics of Selected Leaders

Designated and Elected selection may favor candidates with stronger observed task performance, which could proxy for ability or competence and make them more effective leaders. Consistent with this possibility, candidates with lower addition-task scores receive lower rankings from both members and past leaders under Elected and Designated selection (Table A.3). Selected lead-

ers under these procedures therefore tend to have higher observed scores than candidates who are not selected, with the difference being most pronounced under Elected selection (Table A.4, Panel A).¹³ Within each leader-selection procedure, however, selected leaders' scores do not differ significantly between the Communal and Personal conditions (Table A.4, Panel B). Differences in observed leader performance therefore cannot by themselves explain the interaction between selection procedure and persuasion style.

Table 5: Treatment Effects Controlling for Leader Performance Characteristics

<i>Dep var: Contribution Level</i>	OLS regression		
	(1)	(2)	(3)
Designated	0.078 (0.067)	0.079 (0.066)	0.082 (0.067)
Elected	0.176* (0.100)	0.172 (0.101)	0.175 (0.104)
Random $\times$ COMM	0.190* (0.097)	0.185* (0.097)	0.186* (0.098)
Designated $\times$ COMM	0.151 (0.088)	0.144 (0.086)	0.140 (0.083)
Elected $\times$ COMM	-0.171** (0.076)	-0.177** (0.077)	-0.179** (0.078)
Leader's Score	0.002 (0.001)	0.001 (0.001)	
Leader's Realized Task Difficulty		-0.219 (0.175)	-0.247* (0.141)
Constant	0.219** (0.094)	0.388* (0.189)	0.430*** (0.135)
Round, Period-within-Round Fixed Effects	✓	✓	✓
<i>Differences in Persuasion Effects</i>			
(Random $\times$ COMM – Designated $\times$ COMM)	0.039 (0.109)	0.042 (0.108)	0.046 (0.109)
(Designated $\times$ COMM – Elected $\times$ COMM)	0.322** (0.125)	0.320** (0.126)	0.319** (0.125)
(Random $\times$ COMM – Elected $\times$ COMM)	0.362** (0.135)	0.362** (0.135)	0.365** (0.138)
Observations	8190	8190	8190

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 1 to Period 10 dummies. Realized Task Difficulty is measured as the proportion of question attempts involving two-digit sums. Contribution levels are normalized to [0,1] by dividing contributions by the 20-token endowment.

Table 5 examines this possibility more directly. Because observed scores may reflect both leaders' ability or competence and the difficulty of the questions they encountered, we control for

¹³The smaller difference under Designated selection partly reflects the random selection of leaders by the experimenter in Round 1. The estimated difference becomes larger when these observations are excluded.

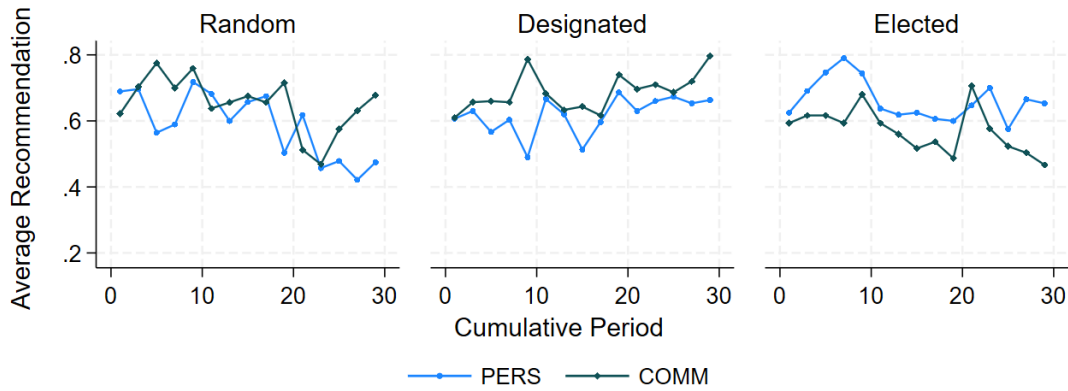
leaders' observed scores, their realized task difficulty, or both. The estimated interaction between leader selection and persuasion style remains quantitatively similar across all three specifications. Observed scores are not significantly associated with members' contributions in Columns 1 and 2, while realized task difficulty is negatively associated with contributions and is marginally significant in Column 3. Taken together, the stability of the interaction across specifications suggests that differences in selected leaders' observed task performance are unlikely to drive our main treatment effects.

Result 2. *Differences in selected leaders' observed task performance do not account for the interaction between leader selection and persuasion style.*

Leaders' behavior

We next examine whether the interaction operates through differences in how leaders exercise their authority. Leader-selection procedures and persuasion styles may affect the recommendations leaders issue or the messages they choose, which could in turn generate differences in member cooperation.¹⁴ We examine contribution recommendations and message choices separately and find little evidence for this channel.

Figure 3: Leaders' Recommendations in Each Period



Recommendations are normalized to the interval $[0,1]$ by dividing recommendations by the 20-token endowment. Because leaders send targets every odd-numbered period, the graph displays only odd-numbered periods.

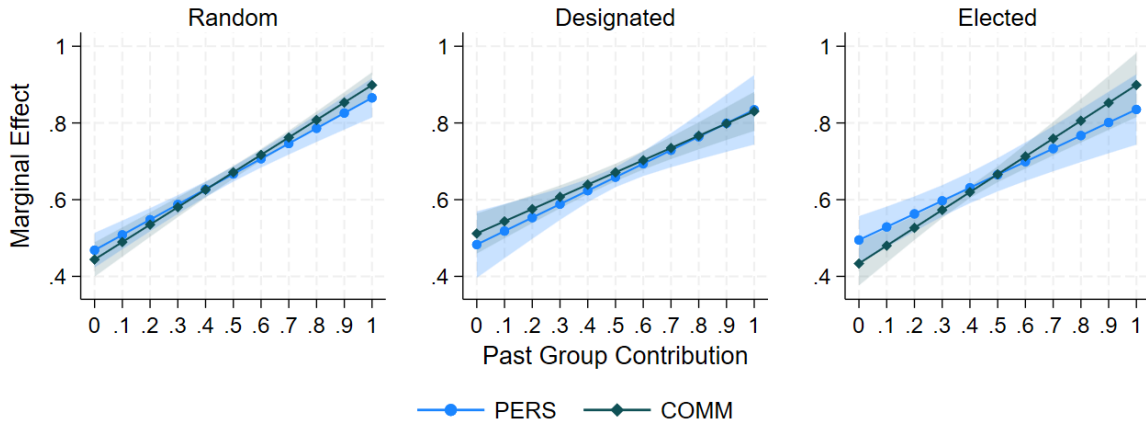
Figure 3 plots leaders' contribution recommendations over time. In the first period, before leaders have observed any decisions by their member-group, recommendations do not differ

¹⁴For example, Elected leaders in the Personal condition may moderate their recommendations or choice of messages because they anticipate greater scrutiny or resistance from members.

across the six treatment conditions (joint test, $p = 0.947$). As interaction proceeds, recommendation patterns begin to resemble those observed for members' contributions in Figure 2, although the differences are less pronounced. These emerging differences, however, appear largely related to leaders' responses to their group's prior behavior, rather than systematic treatment differences in how leaders persuade.

Consistent with this interpretation, Table A.5 shows that treatment differences in recommendations are substantially attenuated after controlling for the member-group's past contribution. We also examine whether leaders differ across treatments in how they adjust their recommendations in response to past contributions, controlling for their past recommendation. Figure 4 shows no significant differences between Communal and Personal persuasion in how past contributions affect recommendations within any leader-selection procedure.

Figure 4: Marginal Effects of Past Group Contributions on Leaders' Recommendations



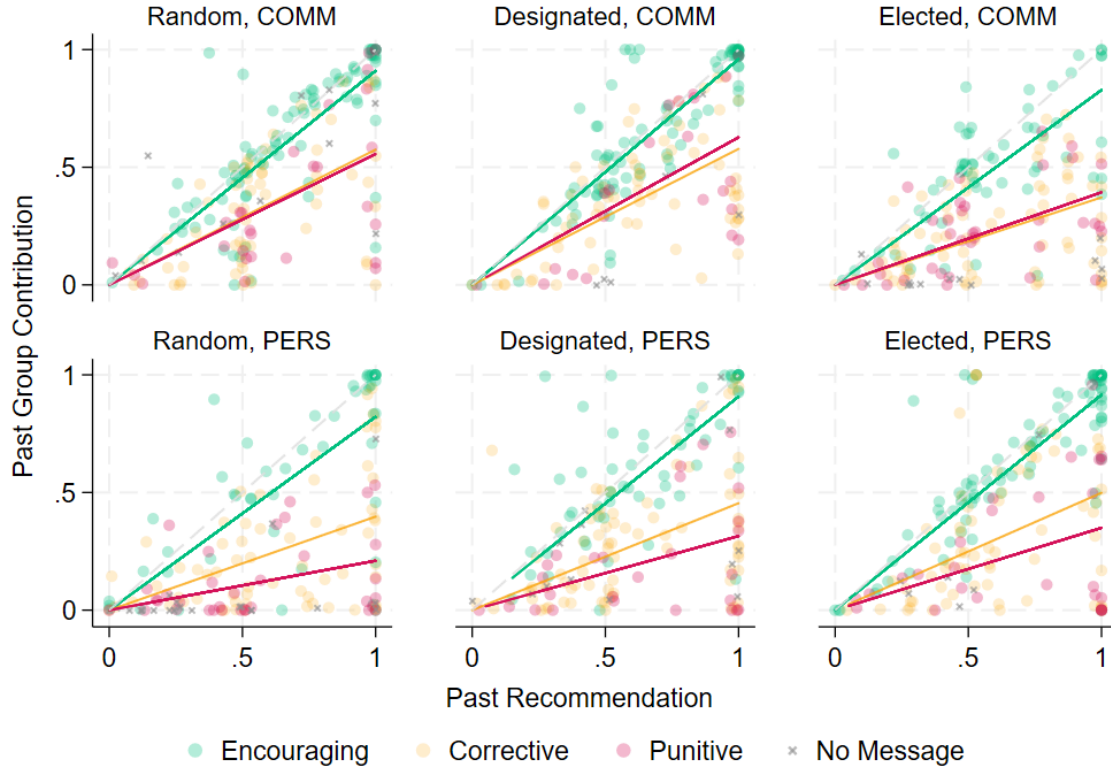
Marginal effects are estimated from coefficients in Table A.6, Column 1, which also reports formal tests of the differences across treatment conditions.

We next examine leaders' message choices.¹⁵ Columns 3–8 of Table A.5 show that, as with recommendations, treatment differences in the use of Encouraging, Corrective, and Punitive messages (if any) largely disappear after controlling for past group contributions. Figure 5 further illustrates that message choices respond strongly to past group behavior. Encouraging messages are used primarily when past contributions are close to the leader's past recommendation, whereas Corrective and Punitive messages become more likely when contributions fall short of that recommendation.

Although message use differs somewhat between Communal and Personal persuasion, these differences do not vary systematically across leader-selection procedures. More directly, Columns 2–

¹⁵Leaders choose to send an additional message in 89.9% of the decision periods in which this option is available.

Figure 5: Message Choices by Past Contributions and Recommendations



Each point represents an observation of a message for a given combination of the member-group's past contribution and the leader's past recommendation, both of which are observed by leaders. Lines of best fit are estimated separately for each message category without a constant.

4 of Table A.6 and Figure A.4 show no significant differences between Communal and Personal persuasion in how leaders' message choices respond to past group contributions within any selection procedure. Taken together, these results provide little evidence that systematic differences in leaders' recommendations or message choices account for the interaction between leader selection and persuasion style. Observed variation in leader behavior is closely related to members' past behavior, while Table A.7 shows that the main treatment interaction remains quantitatively similar when controlling for leaders' recommendations and message choices.

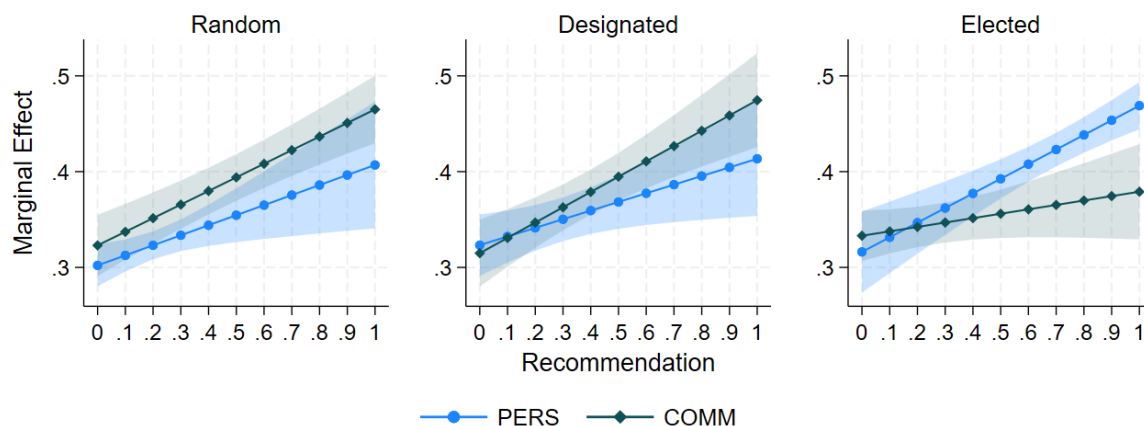
Result 3. *There is little evidence that the interaction between leader selection and persuasion style is driven by differences in how leaders exercise their authority. Treatment differences in leaders' recommendations and message choices largely reflect responses to members' past behavior and disappear once past contributions are controlled for.*

Members' Responses to Leaders

Having found little evidence that the interaction is explained by observed differences in selected leaders or leaders' behavior, we turn to the channel most directly associated with legitimacy: members' willingness to follow leaders' guidance. If leader selection and persuasion style jointly shape whether authority is regarded as legitimate, the same guidance may elicit different responses depending on how the leader was selected and how that guidance was justified.

We examine whether members adjust their contributions differently in response to leaders' recommendations across treatment conditions, controlling for their own past contribution. Figure 6 reports the estimated effect of recommendations on contributions within each condition.

Figure 6: Marginal Effects of Recommendations on Members' Contributions in Each Treatment



Marginal effects are estimated from coefficients in Table A.8, which also shows exact tests of differences.

Under Random selection, Communal persuasion appears to raise members' general willingness to cooperate rather than their responsiveness to the precise recommendation, which does not differ significantly across persuasion styles ($p = 0.483$). Under Designated selection, members are somewhat more responsive to recommendations under Communal persuasion, although the difference is not statistically significant ($p = 0.223$). Under Elected selection, the ordering reverses: members are significantly more responsive to recommendations under Personal than under Communal persuasion ($p = 0.007$). These patterns are broadly consistent with the declining relative effectiveness of Communal persuasion as leader selection becomes more collective. By contrast, there is little evidence that members respond differently across treatment conditions to the category of additional message received (Table A.9).¹⁶

¹⁶As supplementary evidence, Table A.10 examines members' beliefs about others' contributions. The respon-

These differences in members' responses may accumulate over repeated interaction, helping to explain why contributions begin at similar levels in Period 1 but diverge thereafter. Members' evaluations of leader effectiveness also display a qualitatively similar pattern to the contribution pattern in Table 4, although the estimates are less precise (Table A.11).¹⁷

Overall, the mechanism evidence is consistent with leader selection and persuasion style jointly shaping legitimacy and consequently members responses to leaders' guidance. Observable leader characteristics and differences in leaders' recommendations or message choices do not account for the treatment effects. Instead, the interaction is most clearly reflected in members' responses, affecting both their general willingness to cooperate and the weight they place on recommendations.

Result 4. *The interaction is most clearly reflected in members' responses to leaders' guidance. Both their responsiveness to recommendations and their general willingness to cooperate vary with leader selection and persuasion style in a pattern consistent with the treatment effects on contributions.*

4 Conclusion

This paper studies how two sources of leadership legitimacy, leader selection and the justification of leaders' guidance, jointly shape cooperation. In a repeated public-goods game where leaders influence members only through recommendations and fixed-form messages, appeals to shared norms increase cooperation relative to personal appeals when leaders are randomly assigned, but decrease it when leaders are elected. Leader selection and persuasion style therefore cannot be understood independently: their effects depend on how they are combined.

Conceptually, these findings challenge an additive view of collectiveness, under which more collective selection and more collective persuasion should reinforce one another. Instead, selection and persuasion appear to provide alternative bases on which followers accept guidance: each can supply what the other lacks, and once authority rests on a collective basis, further collective

siveness of these beliefs to recommendations follows a qualitatively similar pattern to that of actual contributions: responses differ across persuasion styles depending on the leader-selection procedure. This suggests that leader selection and persuasion style jointly shape not only members' own behavior, but also their expectations about others' behavior.

¹⁷Relative to Personal persuasion, Communal persuasion is associated with more favorable evaluations under Random and Designated selection, but less favorable evaluations under Elected selection.

appeals add little. Legitimacy thus depends not on any single procedure or practice, but on the broader configuration through which authority is conferred, justified, and received.

This interdependence is particularly relevant in practice because leader-selection procedures and persuasion styles are unlikely to be determined independently. Organizational culture and values may shape both how leaders are selected and how they justify their guidance. A culture emphasizing collective participation, for example, may favor both election and appeals to shared values. By contrast, in hierarchical organizations where promotion is based solely on seniority with little bottom-up input, leaders may naturally justify their guidance by appealing to the authority of their position and their personal judgment. In both cases, selection and persuasion draw on similar bases of legitimacy, yet such alignment need not be the most effective. More broadly, our results suggest that organizations should pay greater attention to how different leadership practices interact, rather than evaluating them in isolation.

Several avenues for future research follow. First, future work could examine whether the interaction between leader selection and persuasion style extends to a broader range of incentive and strategic environments, such as settings in which leaders can reward or sanction followers, make binding group decisions, or address coordination problems rather than social dilemmas. Second, allowing leaders to persuade followers through more open-ended communication could reveal how they combine, switch between, or adapt Personal and Communal appeals, as well as how followers' preferences shape the persuasion styles that emerge.¹⁸ Finally, leadership is inherently dynamic. Although our design features repeated interaction, changing leaders, and leader succession, it does not cleanly identify how legitimacy evolves across successive leaders. Future work could examine how past leadership and anticipated succession determine whether legitimacy is transferred, accumulated, or depleted across leaders and institutions.

¹⁸In our post-experimental survey, the share of members preferring Personal rather than Communal appeals is 13.2 percentage points higher under Elected selection than under the other selection procedures ($p = 0.072$). Nevertheless, Communal persuasion remains the majority preference under every selection procedure. This suggests that the styles followers prefer need not coincide with those that are most effective.

Statements and Declarations

The authors have no competing interests in the submission of this article. During the preparation of this work, the authors used ChatGPT to improve grammar and clarity after writing the full draft. All substantive content, argumentation, experimental design, and framing reflect the authors' own thinking. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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A Supplementary Results

Figure A.1: Distribution of Scores for Easy and Hard Versions

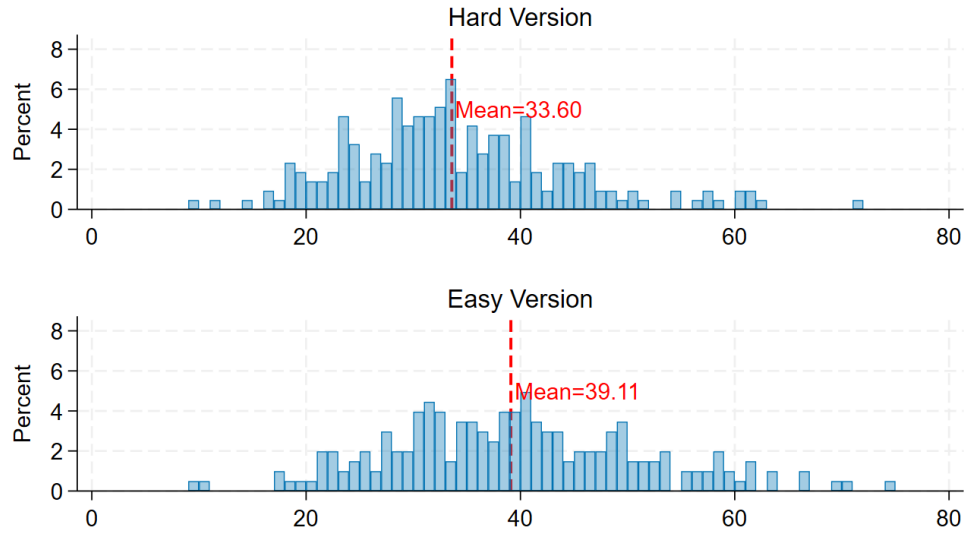
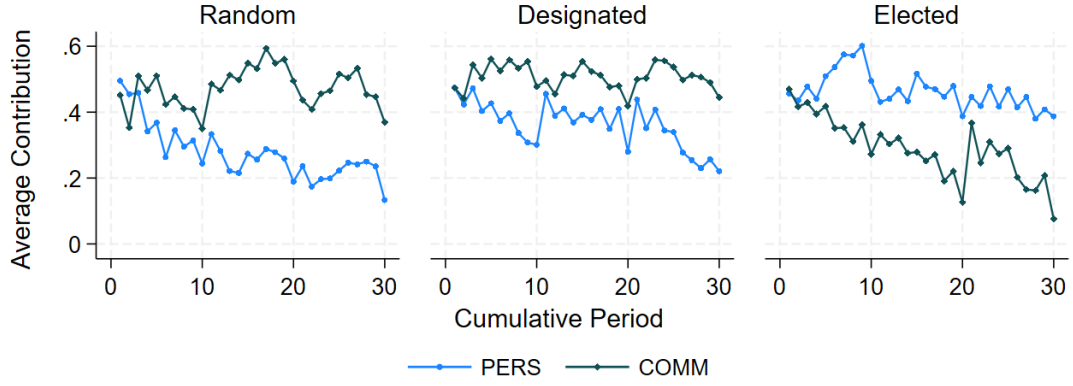


Table A.1: Math Score of Participants

	Candidates	Members	Overall
Average Math Score	36.85	35.96	36.27
	(11.93)	(11.16)	(11.43)

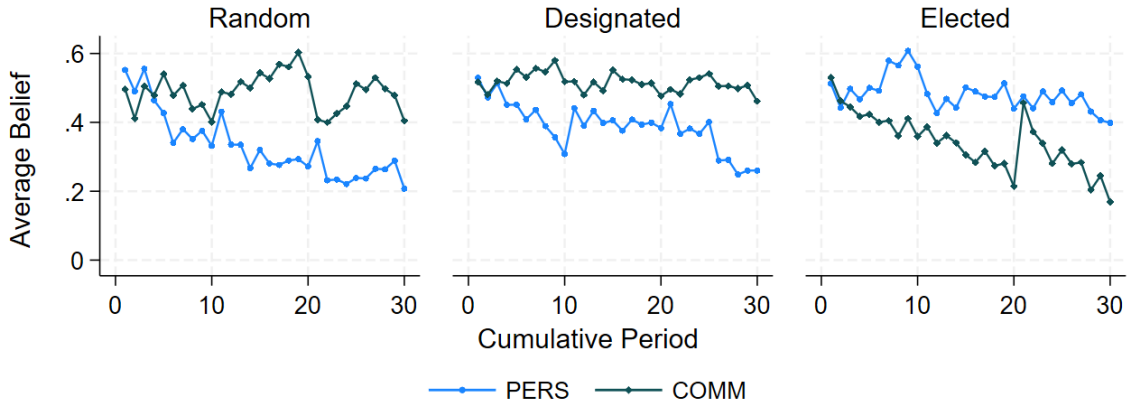
Notes: Means with standard deviation in parentheses.

Figure A.2: Members' Contributions in Each Period



Contribution levels are normalized to $[0,1]$ by dividing contributions by the 20-token endowment.

Figure A.3: Members' Contribution Beliefs in Each Period



Beliefs are normalized to $[0,1]$ by dividing contribution beliefs by the 20-token endowment.

Table A.2: Contribution Beliefs across Treatments

<i>Dep: Contribution Beliefs</i>	(1)	(2)	(3)
Designated	0.043 (0.051)	0.069 (0.066)	0.069 (0.066)
Elected	0.004 (0.079)	0.176* (0.089)	0.176* (0.089)
COMM	0.053 (0.045)		
Random $\times$ COMM		0.184* (0.087)	0.184* (0.087)
Designated $\times$ COMM		0.141* (0.079)	0.141* (0.079)
Elected $\times$ COMM		-0.158** (0.074)	-0.158** (0.074)
Constant	0.379*** (0.060)	0.309*** (0.072)	0.309*** (0.072)
Round, Period-within-Round Fixed Effects			✓
<i>Difference in Persuasion Effects</i>			
(Random $\times$ COMM – Designated $\times$ COMM)		0.043 (0.101)	0.043 (0.101)
(Designated $\times$ COMM – Elected $\times$ COMM)		0.299** (0.117)	0.299** (0.117)
(Random $\times$ COMM – Elected $\times$ COMM)		0.342** (0.127)	0.342** (0.127)
Observations	8190	8190	8190

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 1 to Period 10 dummies. Beliefs are normalized to $[0,1]$ by dividing contribution beliefs by the 20-token endowment.

Table A.3: Determinants of Rankings

<i>Dep Var:</i>	Rank			Was Lowest Rank		
	(1)	(2)	(3)	(4)	(5)	(6)
	Members: All	Members: Elected	Leaders: Designated	Members: All	Members: Elected	Leaders: Designated
Score of Leader	-0.015*** (0.003)	-0.011*** (0.003)	-0.003 (0.004)			
Lowest Score				0.349*** (0.048)	0.312*** (0.043)	0.200*** (0.051)
Constant	2.562*** (0.115)	2.408*** (0.100)	2.111*** (0.132)	0.217*** (0.016)	0.229*** (0.014)	0.267*** (0.017)
Observations	2358	837	180	2358	837	180

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Columns 1, 2, 4, 5: Member-Candidate-Round level OLS regressions with standard errors clustered at the session level in parentheses. Columns 3, 6: Past Leader-Candidate-Round level OLS regressions with standard errors clustered at the session level in parentheses. Ranks range from 1 (first) to 3 (third). Lowest score is a dummy for whether the candidate has the lowest math score in their candidate-pool. Columns 1 and 4 include hypothetical choices of members in Random and Designated conditions.

Table A.4: Performance of Leaders across Leader Selection Conditions

<i>Dep var: Score</i>	All rounds	Excludes first round of Designated
	(1)	(2)
<i>Panel A: Sample of all candidates</i>		
Designated	-2.019 (2.236)	-3.083 (2.175)
Elected	-6.840*** (2.130)	-6.840*** (2.132)
Random $\times$ Leader	-1.389 (2.265)	-1.389 (2.266)
Designated $\times$ Leader	1.841 (1.762)	3.178 (1.850)
Elected $\times$ Leader	9.247*** (1.530)	9.247*** (1.532)
Constant	37.722*** (1.872)	37.722*** (1.873)
Observations	432	384
<i>Panel B: Sample of all leaders</i>		
Designated	4.651** (1.755)	4.662* (2.257)
Elected	6.491*** (1.852)	6.491*** (1.854)
Random $\times$ COMM	2.679 (2.997)	2.679 (3.000)
Designated $\times$ COMM	-4.022 (2.534)	-3.500 (2.994)
Elected $\times$ COMM	-2.618 (2.776)	-2.618 (2.779)
Constant	34.905*** (2.816)	34.905*** (2.819)
Observations	273	243

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Panel A: Candidate-round-level OLS regressions. Panel B: Leader-round-level OLS regressions. Standard errors clustered at the session level in parentheses.

Table A.5: Treatment Effects on Leaders' Messages and Recommendations

<i>Dep var:</i>	Recommendation		Encouraging		Corrective		Punitive	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Designated	0.043 (0.049)	-0.011 (0.031)	0.042 (0.044)	-0.027 (0.032)	0.069 (0.047)	0.107 (0.061)	-0.052 (0.031)	-0.032 (0.027)
Elected	0.098 (0.086)	-0.007 (0.046)	0.188* (0.103)	0.053 (0.054)	-0.027 (0.065)	0.047 (0.062)	-0.062 (0.048)	-0.022 (0.040)
Random $\times$ COMM	0.097 (0.074)	-0.019 (0.025)	0.193** (0.085)	0.046 (0.047)	-0.063 (0.057)	0.017 (0.058)	-0.062 (0.053)	-0.018 (0.044)
Designated $\times$ COMM	0.079 (0.049)	-0.002 (0.031)	0.150 (0.101)	0.046 (0.056)	-0.083 (0.074)	-0.026 (0.049)	-0.028 (0.041)	0.003 (0.035)
Elected $\times$ COMM	-0.112* (0.055)	-0.017 (0.037)	-0.152 (0.096)	-0.031 (0.064)	0.046 (0.068)	-0.020 (0.061)	0.059 (0.038)	0.023 (0.031)
Past Group Contribution		0.601*** (0.037)		0.768*** (0.057)		-0.421*** (0.057)		-0.229*** (0.035)
Constant	0.570*** (0.061)	0.396*** (0.023)	0.286*** (0.073)	0.064* (0.034)	0.381*** (0.042)	0.503*** (0.043)	0.202*** (0.031)	0.268*** (0.030)
Round, Period-within-Round Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
<i>Difference in Persuasion Effects</i>								
(Random $\times$ COMM – Designated $\times$ COMM)	0.017 (0.064)	-0.016 (0.035)	0.043 (0.098)	0.000 (0.052)	0.020 (0.081)	0.044 (0.073)	-0.034 (0.063)	-0.021 (0.056)
(Designated $\times$ COMM – Elected $\times$ COMM)	0.191*** (0.064)	0.015 (0.042)	0.302* (0.160)	0.077 (0.090)	-0.129 (0.114)	-0.006 (0.080)	-0.087 (0.062)	-0.020 (0.047)
(Random $\times$ COMM – Elected $\times$ COMM)	0.208* (0.105)	-0.001 (0.048)	0.345** (0.154)	0.078 (0.086)	-0.109 (0.089)	0.038 (0.080)	-0.121 (0.080)	-0.041 (0.066)
Observations	1092	1092	1092	1092	1092	1092	1092	1092

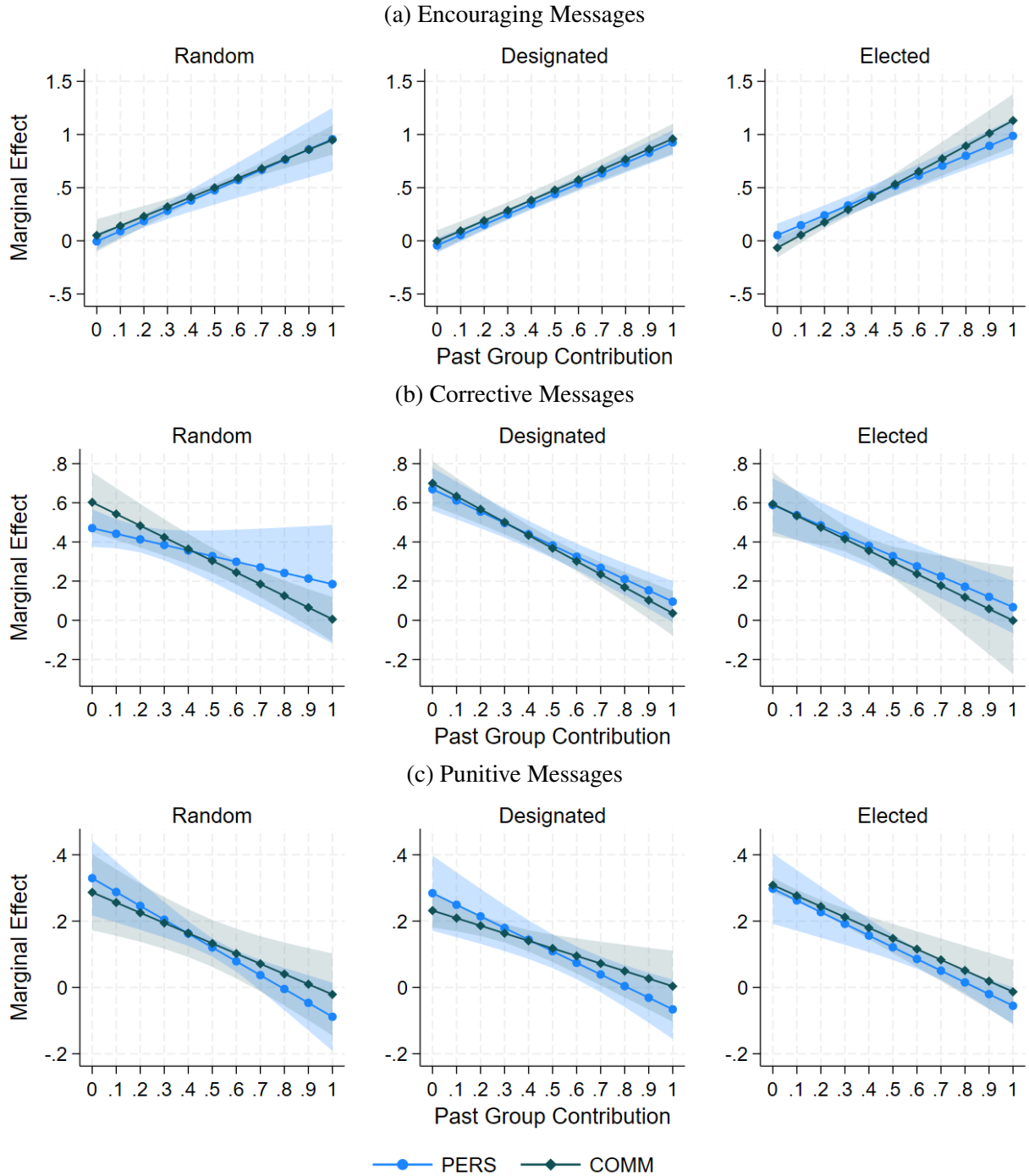
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Leader-odd-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Odd period 3, 5, 7, 9 dummies. The sample includes odd periods where messages are made, and the first period of each block is omitted.

Table A.6: Past Group Contributions and Leaders' Actions: Heterogeneous Effects

<i>Dep var:</i>	Recommendation (1)	Encouraging (2)	Corrective (3)	Punitive (4)
Designated	0.015 (0.035)	-0.038 (0.054)	0.199** (0.073)	-0.045 (0.046)
Elected	0.026 (0.044)	0.058 (0.060)	0.118* (0.058)	-0.032 (0.087)
Random × COMM	-0.024 (0.026)	0.056 (0.087)	0.132 (0.089)	-0.043 (0.075)
Designated × COMM	0.029 (0.042)	0.040 (0.064)	0.030 (0.077)	-0.053 (0.054)
Elected × COMM	-0.061 (0.043)	-0.118 (0.072)	0.004 (0.110)	0.011 (0.059)
Random × PERS × Past Group Contribution	0.397*** (0.045)	0.960*** (0.174)	-0.286 (0.186)	-0.418*** (0.106)
Random × COMM × Past Group Contribution	0.455*** (0.037)	0.897*** (0.143)	-0.597*** (0.120)	-0.308*** (0.098)
Designated × PERS × Past Group Contribution	0.351*** (0.087)	0.967*** (0.077)	-0.574*** (0.087)	-0.350*** (0.090)
Designated × COMM × Past Group Contribution	0.319*** (0.047)	0.961*** (0.092)	-0.664*** (0.106)	-0.228*** (0.074)
Elected × PERS × Past Group Contribution	0.340*** (0.067)	0.934*** (0.099)	-0.522*** (0.083)	-0.353*** (0.077)
Elected × COMM × Past Group Contribution	0.465*** (0.070)	1.196*** (0.160)	-0.595** (0.214)	-0.322*** (0.055)
Past Recommendation	0.426*** (0.057)	-0.395*** (0.073)	0.221*** (0.075)	0.201*** (0.058)
Constant	0.202*** (0.025)	0.243*** (0.059)	0.332*** (0.054)	0.204*** (0.040)
Round, Period-within-Round Fixed Effects	✓	✓	✓	✓
<i>Tests of Differences</i>				
(Random × COMM × Past Grp Contribution — Random × PERS × Past Grp Contribution)	0.058 (0.040)	-0.064 (0.223)	-0.311 (0.219)	0.110 (0.133)
(Designated × COMM × Past Grp Contribution — Designated × PERS × Past Grp Contribution)	-0.033 (0.072)	-0.006 (0.112)	-0.090 (0.118)	0.122 (0.091)
(Elected × COMM × Past Grp Contribution — Elected × PERS × Past Grp Contribution)	0.125 (0.084)	0.263 (0.184)	-0.073 (0.229)	0.031 (0.096)
Observations	1092	1092	1092	1092

* p<0.10, ** p<0.05, *** p<0.01. Leader-odd-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Odd period 3, 5, 7, 9 dummies. The sample includes odd periods where messages and recommendations are made, and the first period of each block is omitted.

Figure A.4: Marginal Effects of Past Group Contributions on Leaders' Messages in Each Treatment



Marginal effects are estimated from coefficients in Table A.6, Columns 2 to 4.

Table A.7: Leaders' Actions and Members' Contributions: Aggregate Effects

<i>Dep Var:</i>	Contributions	
	(1)	(2)
Designated	0.086 (0.069)	0.056 (0.057)
Elected	0.187* (0.104)	0.118 (0.070)
Random $\times$ COMM	0.195* (0.099)	0.131** (0.061)
Designated $\times$ COMM	0.145 (0.087)	0.091 (0.061)
Elected $\times$ COMM	-0.176** (0.076)	-0.108* (0.054)
Recommendation		0.522*** (0.042)
Encouraging Message		0.149*** (0.027)
Corrective Message		-0.050* (0.027)
Punitive Message		-0.106*** (0.026)
Constant	0.277*** (0.083)	-0.029 (0.033)
Round, Period-within-Round Fixed Effects	✓	✓
<i>Difference in Persuasion Effects</i>		
(Random $\times$ COMM – Designated $\times$ COMM)	0.050 (0.110)	0.039 (0.081)
(Designated $\times$ COMM – Elected $\times$ COMM)	0.320** (0.124)	0.199** (0.088)
(Random $\times$ COMM – Elected $\times$ COMM)	0.370** (0.139)	0.238** (0.091)
Observations	8190	8190

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-period level OLS regression with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 1 to Period 10 dummies. Contribution levels and recommendations are normalized to $[0,1]$ by dividing by the 20-token endowment. Recommendations and messages are the same every two periods.

Table A.8: Leaders' Recommendations and Members' Contributions: Heterogeneous Effects

<i>Dep var:</i>	Contributions
Designated	0.021 (0.013)
Elected	0.014 (0.018)
Random $\times$ COMM	0.021 (0.018)
Designated $\times$ COMM	-0.008 (0.021)
Elected $\times$ COMM	0.017 (0.019)
Random $\times$ PERS $\times$ Recommendation	0.105** (0.041)
Random $\times$ COMM $\times$ Recommendation	0.142*** (0.024)
Designated $\times$ PERS $\times$ Recommendation	0.090** (0.040)
Designated $\times$ COMM $\times$ Recommendation	0.160*** (0.035)
Elected $\times$ PERS $\times$ Recommendation	0.153*** (0.028)
Elected $\times$ COMM $\times$ Recommendation	0.046 (0.031)
Past Contribution	0.787*** (0.024)
Constant	-0.018 (0.011)
Round, Period-within-Round Fixed Effects	✓
<i>Tests of Differences</i>	
(Random $\times$ COMM $\times$ Recommendation – Random $\times$ PERS $\times$ Recommendation)	0.037 (0.052)
(Designated $\times$ COMM $\times$ Recommendation – Designated $\times$ PERS $\times$ Recommendation)	0.070 (0.055)
(Elected $\times$ COMM $\times$ Recommendation – Elected $\times$ PERS $\times$ Recommendation)	-0.107*** (0.034)
Observations	7371

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-period level OLS regression with standard errors clustered at the session level. Excludes the first period of each round due to a lack of past Contribution data. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 2 to Period 10 dummies.

Table A.9: Leaders' Messages and Members' Contributions: Heterogeneous Effects

<i>Dep var: Contributions</i>	Message Type:		
	Encouraging (1)	Corrective (2)	Punitive (3)
Designated	0.002 (0.011)	0.021 (0.023)	0.016 (0.016)
Elected	0.035 (0.023)	0.048 (0.028)	0.051* (0.025)
Random $\times$ COMM	0.039* (0.019)	0.051* (0.027)	0.045 (0.026)
Designated $\times$ COMM	0.036** (0.016)	0.034 (0.031)	0.042 (0.025)
Elected $\times$ COMM	-0.041** (0.014)	-0.056** (0.020)	-0.052*** (0.016)
Random $\times$ PERS $\times$ Message	0.044** (0.018)	-0.017 (0.015)	-0.007 (0.028)
Random $\times$ COMM $\times$ Message	0.048*** (0.014)	-0.042*** (0.012)	-0.017 (0.020)
Designated $\times$ PERS $\times$ Message	0.073*** (0.018)	-0.035* (0.019)	-0.037 (0.023)
Designated $\times$ COMM $\times$ Message	0.068*** (0.021)	-0.030 (0.020)	-0.059*** (0.020)
Elected $\times$ PERS $\times$ Message	0.061*** (0.019)	-0.023 (0.014)	-0.047** (0.017)
Elected $\times$ COMM $\times$ Message	0.041** (0.018)	-0.007 (0.013)	-0.029*** (0.007)
Past Contribution	0.810*** (0.020)	0.835*** (0.020)	0.837*** (0.019)
Constant	0.024* (0.012)	0.035** (0.015)	0.030 (0.018)
Round, Period-within-Round Fixed Effects	✓	✓	✓
<i>Tests of Differences</i>			
(Random $\times$ COMM $\times$ Message – Random $\times$ PERS $\times$ Message)	0.004 (0.023)	-0.025 (0.020)	-0.010 (0.034)
(Designated $\times$ COMM $\times$ Message – Designated $\times$ PERS $\times$ Message)	-0.004 (0.029)	0.006 (0.029)	-0.022 (0.031)
(Elected $\times$ COMM $\times$ Message – Elected $\times$ PERS $\times$ Message)	-0.020 (0.024)	0.016 (0.018)	0.019 (0.019)
Observations	7371	7371	7371

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-period level OLS regressions with standard errors clustered at the session level. Excludes the first period of each round due to a lack of past Contribution data. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 2 to Period 10 dummies.

Table A.10: Leaders' Recommendations and Members' Beliefs: Heterogeneous Effects

<i>Dep Var:</i>	Contribution Belief
Designated	0.021 (0.023)
Elected	0.003 (0.025)
Random $\times$ COMM	-0.009 (0.016)
Designated $\times$ COMM	-0.024 (0.026)
Elected $\times$ COMM	0.003 (0.024)
Random $\times$ PERS $\times$ Recommendation	0.148*** (0.038)
Random $\times$ COMM $\times$ Recommendation	0.251*** (0.030)
Designated $\times$ PERS $\times$ Recommendation	0.128** (0.053)
Designated $\times$ COMM $\times$ Recommendation	0.234*** (0.037)
Elected $\times$ PERS $\times$ Recommendation	0.225*** (0.040)
Elected $\times$ COMM $\times$ Recommendation	0.156*** (0.041)
Past Contribution	0.661*** (0.024)
Constant	0.024* (0.012)
Round, Period-within-Round Fixed Effects	✓
<i>Tests of Differences</i>	
(Random $\times$ COMM $\times$ Recommendation – Random $\times$ PERS $\times$ Recommendation)	0.103** (0.043)
(Designated $\times$ COMM $\times$ Recommendation – Designated $\times$ PERS $\times$ Recommendation)	0.106 (0.064)
(Elected $\times$ COMM $\times$ Recommendation – Elected $\times$ PERS $\times$ Recommendation)	-0.070 (0.051)
Observations	7371

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-period level OLS regression with standard errors clustered at the session level in parentheses. The first period of each round is excluded due to a lack of past contribution data. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 2 to Period 10 dummies. Beliefs about other members' contributions and recommendations are normalized to [0,1] by dividing by the 20-token endowment.

Table A.11: Members' Perceptions of leaders

<i>Dep Var:</i>	Leader was important (1)	Leader's message mattered (2)	Rating of leader (3)
Designated	0.163 (0.197)	0.353 (0.223)	-0.048 (0.181)
Elected	0.375 (0.346)	0.600* (0.324)	0.063 (0.222)
Random $\times$ COMM	0.694* (0.374)	0.628 (0.374)	0.194 (0.268)
Designated $\times$ COMM	0.378 (0.268)	0.326 (0.303)	0.215 (0.222)
Elected $\times$ COMM	-0.331 (0.248)	-0.402 (0.233)	-0.237 (0.161)
Constant	2.000*** (0.249)	2.143*** (0.221)	2.944*** (0.168)
Round Fixed Effects	✓	✓	✓
Observations	819	819	819

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Member-round level OLS regression with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. All dependent variables are on a 1-5 Likert scale, where a higher number reflects a more positive opinion. See Appendix D.3 for more details.

B Robustness

This section replicates Table 4, Table 5, and Tables A.5–A.8 using wild-cluster bootstrap p -values to assign significance stars. In every replication, the reported coefficients and standard errors are based on the original specification with standard errors clustered at the session level.

Table B.1: Wild-Bootstrap Replication of Table 4: Cooperation across Treatments

<i>Dep: Contribution Level</i>	(1)	(2)	(3)
Designated	0.056 (0.056)	0.086 (0.069)	0.086 (0.069)
Elected	0.000 (0.085)	0.187 (0.104)	0.187 (0.104)
COMM	0.052 (0.050)		
Random $\times$ COMM		0.195* (0.099)	0.195* (0.099)
Designated $\times$ COMM		0.145 (0.087)	0.145 (0.087)
Elected $\times$ COMM		-0.176** (0.076)	-0.176** (0.076)
Constant	0.353*** (0.067)	0.277*** (0.083)	0.277*** (0.083)
Round, Period-within-Round Fixed Effects			✓
<i>Difference in Persuasion Effects</i>			
(Random $\times$ COMM – Designated $\times$ COMM)		0.050 (0.110)	0.050 (0.110)
(Designated $\times$ COMM – Elected $\times$ COMM)		0.320** (0.124)	0.320** (0.124)
(Random $\times$ COMM – Elected $\times$ COMM)		0.370** (0.139)	0.370** (0.139)
Observations	8190	8190	8190

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Significance stars are based on wild-cluster bootstrap p -values using 10,000 replications, Webb weights. Member-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 1 to Period 10 dummies. Contribution levels are normalized to $[0,1]$ by dividing contributions by the 20-token endowment.

Table B.2: Wild-Bootstrap Replication of Table 5: Treatment Effects Controlling for Leader Characteristics

<i>Dep var: Contribution Level</i>	OLS regression		
	(1)	(2)	(3)
Designated	0.078 (0.067)	0.079 (0.066)	0.082 (0.067)
Elected	0.176 (0.100)	0.172 (0.101)	0.175 (0.104)
Random $\times$ COMM	0.190* (0.097)	0.185* (0.097)	0.186* (0.098)
Designated $\times$ COMM	0.151 (0.088)	0.144 (0.086)	0.140 (0.083)
Elected $\times$ COMM	-0.171* (0.076)	-0.177** (0.077)	-0.179** (0.078)
Leader's Score	0.002 (0.001)	0.001 (0.001)	
Leader's Realized Task Difficulty		-0.219 (0.175)	-0.247 (0.141)
Constant	0.219** (0.094)	0.388* (0.189)	0.430*** (0.135)
Round, Period-within-Round Fixed Effects	✓	✓	✓
<i>Difference in Persuasion Effects</i>			
(Random $\times$ COMM – Designated $\times$ COMM)	0.039 (0.109)	0.042 (0.108)	0.046 (0.109)
(Designated $\times$ COMM – Elected $\times$ COMM)	0.322** (0.125)	0.320** (0.126)	0.319** (0.125)
(Random $\times$ COMM – Elected $\times$ COMM)	0.362** (0.135)	0.362** (0.135)	0.365** (0.138)
Observations	8190	8190	8190

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Significance stars are based on wild-cluster bootstrap p -values using 10,000 replications, Webb weights. Member-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 1 to Period 10 dummies. Realized Task Difficulty is measured as the proportion of question attempts involving two-digit sums. Contribution levels are normalized to $[0,1]$ by dividing contributions by the 20-token endowment.

Table B.3: Wild-Bootstrap Replication of Table A.5: Treatment Effects on Leaders' Messages and Recommendations

<i>Dep var:</i>	Recommendation		Encouraging		Corrective		Punitive	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Designated	0.043 (0.049)	-0.011 (0.031)	0.042 (0.044)	-0.027 (0.032)	0.069 (0.047)	0.107 (0.061)	-0.052 (0.031)	-0.032 (0.027)
Elected	0.098 (0.086)	-0.007 (0.046)	0.188 (0.103)	0.053 (0.054)	-0.027 (0.065)	0.047 (0.062)	-0.062 (0.048)	-0.022 (0.040)
Random $\times$ COMM	0.097 (0.074)	-0.019 (0.025)	0.193* (0.085)	0.046 (0.047)	-0.063 (0.057)	0.017 (0.058)	-0.062 (0.053)	-0.018 (0.044)
Designated $\times$ COMM	0.079 (0.049)	-0.002 (0.031)	0.150 (0.101)	0.046 (0.056)	-0.083 (0.074)	-0.026 (0.049)	-0.028 (0.041)	0.003 (0.035)
Elected $\times$ COMM	-0.112* (0.055)	-0.017 (0.037)	-0.152 (0.096)	-0.031 (0.064)	0.046 (0.068)	-0.020 (0.061)	0.059 (0.038)	0.023 (0.031)
Past Group Contribution		0.601*** (0.037)		0.768*** (0.057)		-0.421*** (0.057)		-0.229*** (0.035)
Constant	0.570*** (0.061)	0.396*** (0.023)	0.286*** (0.073)	0.064 (0.034)	0.381*** (0.042)	0.503*** (0.043)	0.202*** (0.031)	0.268*** (0.030)
Round, Period-within-Round Fixed Effects	✓	✓	✓	✓	✓	✓	✓	✓
<i>Difference in Persuasion Effects</i>								
(Random $\times$ COMM– Designated $\times$ COMM)	0.017 (0.064)	-0.016 (0.035)	0.043 (0.098)	0.000 (0.052)	0.020 (0.081)	0.044 (0.073)	-0.034 (0.063)	-0.021 (0.056)
(Designated $\times$ COMM– Elected $\times$ COMM)	0.191** (0.064)	0.015 (0.042)	0.302* (0.160)	0.077 (0.090)	-0.129 (0.114)	-0.006 (0.080)	-0.087 (0.062)	-0.020 (0.047)
(Random $\times$ COMM– Elected $\times$ COMM)	0.208* (0.105)	-0.001 (0.048)	0.345* (0.154)	0.078 (0.086)	-0.109 (0.089)	0.038 (0.080)	-0.121 (0.080)	-0.041 (0.066)
Observations	1092	1092	1092	1092	1092	1092	1092	1092

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Significance stars are based on wild-cluster bootstrap p -values using 10,000 replications, Webb weights. Leader-odd-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Odd period 3, 5, 7, 9 dummies. The sample includes odd periods where messages are made, and the first period of each block is omitted.

Table B.4: Wild-Bootstrap Replication of Table A.6: Past Group Contributions and Leaders' Actions

<i>Dep var:</i>	Recommendation (1)	Encouraging (2)	Corrective (3)	Punitive (4)
Designated	0.015 (0.035)	-0.038 (0.054)	0.199* (0.073)	-0.045 (0.046)
Elected	0.026 (0.044)	0.058 (0.060)	0.118 (0.058)	-0.032 (0.087)
Random $\times$ COMM	-0.024 (0.026)	0.056 (0.087)	0.132 (0.089)	-0.043 (0.075)
Designated $\times$ COMM	0.029 (0.042)	0.040 (0.064)	0.030 (0.077)	-0.053 (0.054)
Elected $\times$ COMM	-0.061 (0.043)	-0.118 (0.072)	0.004 (0.110)	0.011 (0.059)
Random $\times$ PERS $\times$ Past Group Contribution	0.397*** (0.045)	0.960*** (0.174)	-0.286 (0.186)	-0.418** (0.106)
Random $\times$ COMM $\times$ Past Group Contribution	0.455*** (0.037)	0.897*** (0.143)	-0.597** (0.120)	-0.308*** (0.098)
Designated $\times$ PERS $\times$ Past Group Contribution	0.351** (0.087)	0.967*** (0.077)	-0.574*** (0.087)	-0.350** (0.090)
Designated $\times$ COMM $\times$ Past Group Contribution	0.319*** (0.047)	0.961*** (0.092)	-0.664*** (0.106)	-0.228* (0.074)
Elected $\times$ PERS $\times$ Past Group Contribution	0.340*** (0.067)	0.934*** (0.099)	-0.522*** (0.083)	-0.353*** (0.077)
Elected $\times$ COMM $\times$ Past Group Contribution	0.465** (0.070)	1.196*** (0.160)	-0.595* (0.214)	-0.322*** (0.055)
Past Recommendation	0.426*** (0.057)	-0.395*** (0.073)	0.221*** (0.075)	0.201*** (0.058)
Constant	0.202*** (0.025)	0.243*** (0.059)	0.332*** (0.054)	0.204*** (0.040)
Round, Period-within-Round Fixed Effects	✓	✓	✓	✓
<i>Tests of Differences</i>				
(Random $\times$ COMM $\times$ Past Grp Contribution — Random $\times$ PERS $\times$ Past Grp Contribution)	0.058 (0.040)	-0.064 (0.223)	-0.311 (0.219)	0.110 (0.133)
(Designated $\times$ COMM $\times$ Past Grp Contribution — Designated $\times$ PERS $\times$ Past Grp Contribution)	-0.033 (0.072)	-0.006 (0.112)	-0.090 (0.118)	0.122 (0.091)
(Elected $\times$ COMM $\times$ Past Grp Contribution — Elected $\times$ PERS $\times$ Past Grp Contribution)	0.125 (0.084)	0.263 (0.184)	-0.073 (0.229)	0.031 (0.096)
Observations	1092	1092	1092	1092

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Significance stars are based on wild-cluster bootstrap p -values using 10,000 replications, Webb weights. Leader-odd-period level OLS regressions with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Odd period 3, 5, 7, 9 dummies. The sample includes odd periods where messages and recommendations are made, and the first period of each block is omitted.

Table B.5: Wild-Bootstrap Replication of Table A.7: Leaders' Actions and Members' Contributions

<i>Dep Var:</i>	Contributions	
	(1)	(2)
Designated	0.086 (0.069)	0.056 (0.057)
Elected	0.187 (0.104)	0.118 (0.070)
Random $\times$ COMM	0.195* (0.099)	0.131* (0.061)
Designated $\times$ COMM	0.145 (0.087)	0.091 (0.061)
Elected $\times$ COMM	-0.176** (0.076)	-0.108* (0.054)
Recommendation		0.522*** (0.042)
Encouraging Message		0.149*** (0.027)
Corrective Message		-0.050* (0.027)
Punitive Message		-0.106*** (0.026)
Constant	0.277*** (0.083)	-0.029 (0.033)
Round, Period-within-Round Fixed Effects	✓	✓
<i>Difference in Difference</i>		
(Random $\times$ COMM – Designated $\times$ COMM)	0.050 (0.110)	0.039 (0.081)
(Designated $\times$ COMM – Elected $\times$ COMM)	0.320** (0.124)	0.199** (0.088)
(Random $\times$ COMM – Elected $\times$ COMM)	0.370** (0.139)	0.238** (0.091)
Observations	8190	8190

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Significance stars are based on wild-cluster bootstrap p -values using 10,000 replications, Webb weights. Member-period level OLS regression with standard errors clustered at the session level in parentheses. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 1 to Period 10 dummies. Contribution levels and recommendations are normalized to $[0,1]$ by dividing by the 20-token endowment. Recommendations and messages are the same every two periods.

Table B.6: Wild-Bootstrap Replication of Table A.8: Leaders' Recommendations and Members' Contributions

<i>Dep var:</i>	Contributions
Designated	0.021 (0.013)
Elected	0.014 (0.018)
Random $\times$ COMM	0.021 (0.018)
Designated $\times$ COMM	-0.008 (0.021)
Elected $\times$ COMM	0.017 (0.019)
Random $\times$ PERS $\times$ Recommendation	0.105** (0.041)
Random $\times$ COMM $\times$ Recommendation	0.142*** (0.024)
Designated $\times$ PERS $\times$ Recommendation	0.090* (0.040)
Designated $\times$ COMM $\times$ Recommendation	0.160*** (0.035)
Elected $\times$ PERS $\times$ Recommendation	0.153*** (0.028)
Elected $\times$ COMM $\times$ Recommendation	0.046 (0.031)
Past Contribution	0.787*** (0.024)
Constant	-0.018 (0.011)
Round, Period-within-Round Fixed Effects	✓
<i>Tests of Differences</i>	
(Random $\times$ COMM $\times$ Recommendation – Random $\times$ PERS $\times$ Recommendation)	0.037 (0.052)
(Designated $\times$ COMM $\times$ Recommendation – Designated $\times$ PERS $\times$ Recommendation)	0.070 (0.055)
(Elected $\times$ COMM $\times$ Recommendation – Elected $\times$ PERS $\times$ Recommendation)	-0.107*** (0.034)
Observations	7371

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Significance stars are based on wild-cluster bootstrap p -values using 10,000 replications, Webb weights. Member-period level OLS regression with standard errors clustered at the session level. Excludes the first period of each round due to a lack of past Contribution data. Round fixed effects: Round 1 to 3 dummies. Period-within-Round fixed effects: Period 2 to Period 10 dummies.

Online appendix to “The Bases of Legitimacy: How Leaders Are Chosen Shapes How They Should Persuade”

C GABRIEL Classification of Messages

To classify the communicative content of the additional leader messages, we follow the logic of the GABRIEL framework proposed by [Asirvatham et al. \(2026\)](#). GABRIEL treats qualitative material, such as text, images, or audio, as data that can be converted into structured variables through GPT prompts. This method requires the researcher to first define the construct of interest in natural language, and GPT then applies the definitions to classify observations. In this methodology, GPT is only used as a measurement tool rather than as a source of categories.

Following the GABRIEL logic, we pre-specified three mutually exclusive substantive categories and their definitions as follows:

Encouraging: “The message uses positive reinforcement, praise, or encouragement. It emphasizes previous success, sustaining a good outcome, or doing well.”

Corrective: “The message indicates that the current behavior or trend should be changed. It asks the member or group to adjust course, invest more, or return to a desired standard.”

Punitive: “The message uses negative pressure, blame, shame, or threat. It suggests that the member should feel ashamed, has violated expectations, or may be excluded from the group.”

The classification prompt instructed the model to assign each message to one of the categories. The Communal and Personal versions of each message were examined separately, allowing us to verify that the classification captured the common communicative strategy rather than being contaminated by the authoritative source embedded in the wording. The categories were specified as mutually exclusive, and the model was instructed to base its classification only on the content of the message, without using the message identifier. For each message, the classification procedure was repeated ten times for both gpt-5.4 and gpt-5.5. The final classification reported in [Table 2](#) is based on the modal category across repetitions. The classifications are stable across repeated runs and consistent across the two models.

D Experimental Materials

D.1 Instructions

General Instructions

Ground Rules

Welcome to the study. Please read the instructions below carefully.

Communication between participants is not allowed. Also, please refrain from using any communication devices. Participants intentionally violating the rules may be asked to leave the experiment and may not be paid. If you have any questions at any time, please raise your hand and an experimenter will come over to see you.

If you need to write anything, please use the paper and pen provided. Please do not write anything on this instruction sheet.

General information

This study will consist of completing two tasks. You will receive instructions on each task before they start. At the end of the study, you will also be asked to fill in a questionnaire.

Your decisions in the study will be anonymous, and your anonymity will be strictly preserved. Any interactions with other participants will only be via computer assigned unique IDs which will be used for the whole duration of the study.

Your total earnings will be equal to the sum of the earnings from both tasks. All earnings during the study are denominated in tokens. Your tokens will be converted into Singapore dollars at the rate of **20 tokens = \$1**. You will also receive a completion fee of **\$3**. You will be paid privately and confidentially.

Addition Task

Details of the addition task

Now, you will be required to complete a mathematical addition task. You will be given two random numbers (up to 2 digits) and asked to find their mathematical sum.



The screenshot shows a user interface for an addition task. At the top, it displays 'Time left: 118'. Below that, it shows 'Score: 0'. The main part of the interface contains the equation '2 + 0 =', followed by a blue rectangular input box for the answer. At the bottom, there is a button labeled 'Submit (Next question)'.

As shown in the screenshot, solve the sum of the two numbers, fill in your answer in the given blank and click submit to proceed to the next question. Note that blank answers cannot be submitted.

If your answer is correct, your displayed score will increase by 1.

If your answer is incorrect, your displayed score will remain the same.

You will have 2 minutes to solve as many questions as you can.

Earnings in addition task

You will be paid **0.5 tokens** for each question you solve correctly.

I.e. Your earnings in the decision task in tokens is hence equal to **your score** $\times$ 0.5.

Decision Task

Now, you will have to complete a decision task together with others in a group over a number of rounds. Depending on your own and other participants' actions, you can earn a considerable amount of money. It is therefore very important that you understand the following instructions.

Roles

There are two roles in the decision task. You will keep the same role throughout.

At the beginning of the decision task, the computer will randomly assign you either to be 1) a **Candidate**, or 2) a **Member**. Candidates have the chance to become a **Leader** of a group of members.

Both candidates and members will be assigned a unique ID which they will keep throughout.



Candidates will be assigned IDs which are colours like "Red", "Blue" etc.



Members will be assigned IDs which are fruits like "Mango", "Apple" etc.

Rounds and Periods

You will have to complete a decision task over **several rounds**.

Each round of the decision task will consist of **10 periods**.

In each period, members in a group will be involved in investing in a joint project. Leaders will be given the opportunity, in some periods, to communicate with members in the group before the members invest. At the end of the period, all information about investments is revealed to the group.

Groups and Pools

All *members* will be randomly assigned by the computer into **groups** of 3. The members in each group will remain fixed in all rounds.

All *candidates* will also be randomly assigned by the computer into **pools** of 3. The candidates in each pool will also remain fixed in all rounds.

Assigning of Leaders to groups of Members

Each group of 3 members will have a leader to provide guidance during the decision task. The leader assigned will change every round of 10 periods.

At the beginning of each round of 10 periods, each group will be matched with a new pool of candidates. Groups will be assigned a leader from the pool of candidates based on an assignment procedure. The exact assignment procedure will be described in greater detail during the actual task.

Note that not all leader candidates will be assigned to a group.

Details of the Decision Task

1. Members



In every period, each member in the group will receive 20 tokens, and must decide how many of these tokens to keep for themselves, and how many to invest in a joint project.

The task earnings of a member in **each period** consists of

1. Tokens kept by the member for themselves
2. Income from the project = $0.5 \times (\text{total investment by all members in the group})$

Therefore, if as a member, you invest X tokens, while the other two members invest Y and Z tokens respectively, you will earn: $(20 - X) + 0.5 \times (X + Y + Z)$ tokens in that period.

2. Leaders



Every odd-numbered period (i.e., periods 1, 3, 5, 7, 9), the leader of a group can send a single public message to all members in the group. **Members will have to read the leader's message before deciding on their investments.** The interface provides the leader with guidelines for the message and restricts what the leader can say.

The task earnings of a leader in **each period** consists of

1. A fixed payment of 15 tokens
2. Income from the project = $0.2 \times (\text{total investment by all members in the group})$.

Therefore, if the 3 members in the group invest X , Y and Z tokens respectively, the leader will earn $15 + 0.2 \times (X + Y + Z)$ tokens in that period.

3. Non-assigned Candidates



Candidates in the pool not assigned to a matched group in any particular round will instead “shadow” it. “Shadow leaders” do not make any actual decisions, but will observe all the contributions and messages sent in their shadowed group. They will be asked to send **hypothetical messages** in odd-numbered periods. These messages will not be received by members of the group.

The task earnings of a “shadow leader” in **each period** consists of

1. A fixed payment of 10 tokens
2. Income from the project = $0.2 \times (\text{total investment by all members in the group})$.

Therefore, if the 3 members in the group invest X , Y and Z tokens respectively, the “shadow leader” will earn $10 + 0.2 \times (X + Y + Z)$ tokens in that period.

4. Investment Information

At the end of each period, information on each member's investments will be displayed to all group members, the group leader (and shadow leaders if any).

5. Beliefs

Everyone will also be asked their beliefs about others' average contributions.

- Leaders and Shadow Leaders will be asked how much they believe members of the group will invest on average.
- Members will be asked how much they believe other members in the group (not including themselves) will invest on average.

This information will be kept private. You will receive a prediction earning of 0.5 tokens if your answer is within 2 tokens of the correct answer.

6. Round Payoff

The total payoff in a round is equal to the sum of the task and prediction earnings for the round.

Earnings in the Decision Task

Your earnings (either as a member, leader, or non-assigned candidate) will be based on the sum of tokens earned in **3 randomly selected periods from each round**.

To ensure that you understand the instructions, there will be a quiz about the decision task and a practice round. You will also play an anonymous round before we begin the decision task.

Practice Round

In the practice round, you will have the opportunity to experience the user interface of the decision task. You will be interacting with simulated others as a member.

All IDs you see during the practice round will be tentative. Only during the start of the actual rounds of the decision task, will you receive your unique ID.

Anonymous Round

After the practice round and before starting the actual rounds of the decision task, you will also play — as a member — a *similar version* of the above decision task, without any leaders.

There will only be 1 period in this round and everyone's interactions in this round will be anonymous.

You will be matched with two other randomly assigned participants and be asked how much you want to invest in a joint project.

Earnings in Anonymous Round

Your task earnings here will be calculated based on your investment and the investment of the other two members in the single period, like in the standard decision task.

You will also receive a prediction earning of 0.5 tokens if your answer is within 2 tokens of the correct answer.

You will only learn of the exact amount earned here at the end of the experiment.

D.2 Screenshots

(I) Stage 1: Addition Task

Addition task

Time left: 116

Score: 0

1 + 25 =

Addition task

You have completed the addition task.

Your total score in the addition task is 1.
Your total earnings for stage 1 is 0.5 token(s).

You will now move on to the second task which is a decision task.
The experimenter will now hand out instructions for the decision task.

(II) Practice Round

Practice Round

Congratulations on completing the control quiz.

You will now have a practice round of the decision task.

You will play 2 consecutive periods of the decision task as a member. There will be no leader assignment stage in this practice round.

Note that your ID in this round is **temporary**. You will be assigned your actual ID at the start of the actual decision task.

Also note that other members and leaders are **simulated** for this practice round. You will **not** be interacting with actual participants.

Please click proceed to begin.

Proceed

Decision task

Messages can be sent by your leader in this period.

Your leader is currently deciding on the public message to send to you and your group members.

Type NEXT in the box below and click submit to continue

NEXT

Submit

Round: Practice

Period: 1

Your Information



Role: Member

ID: Pomelo

Group Information

Leader: Grey (Com)

Group members: Pomelo
Pear (Com)
Lime (Com)

Round: Practice
Period: 1

Your Information

Role: Member
ID: Pomelo

Group Information
Leader: Grey (Com)
Group members: Pomelo, Pear (Com), Lime (Com)

Decision task

Your leader, Computer Grey, has sent you the following message:

"SAMPLE MESSAGE X"

You have received an endowment of 20 tokens for this period.

Please enter the number of tokens you would like to invest in the joint project.

Your investment in the project:

Please also guess the average number of tokens the other members in your group will invest in this period.

Other members' average investment:

Submit

Round: Practice
Period: 1

Your Information

Role: Member
ID: Pomelo

Group Information
Leader: Grey (Com)
Group members: Pomelo, Pear (Com), Lime (Com)

Decision task - summary

Below is a summary of the contributions all followers have made during this period.

Member	Investment
Pomelo (you)	0
Computer Pear	11
Computer Lime	12

You invested 0 token(s) in the joint project and kept 20 token(s) for yourself.
A total of 23 token(s) were invested in the joint project.
Your earnings for the decision task is 31.5 token(s).

Your prediction on the average contribution of others is 10.0 token(s).
The actual average contribution of others is 11.5 token(s).
The reward from your prediction is 0.5 token(s).

Proceed

Explanation
Your Task Earnings

- Tokens kept for yourself + 0.5 x total tokens invested
- 20 + 0.5x23
- 31.5

Your Prediction Earnings

- 0.5

Because your prediction was within 2 tokens of the actual average contribution of others.

Leader's Task Earnings

- 15 + 0.2 x total tokens invested
- 15 + 0.2x23
- 19.6

Leader's Prediction Earnings

The leader's prediction earnings will depend on whether their prediction is within 2 tokens of the actual average contribution of the group.

(III) Anonymous Round

Anonymous Round

You are now matched with two other randomly assigned participants.
Everyone's interactions in this round will be anonymous.

You have received an endowment of 20 tokens for this period.

Please enter the number of tokens you would like to invest in the joint project.

Your investment in the project:

Please also guess the average number of tokens the other members in your group will invest in this period.

Other members' average investment:

Anonymous Round End

You have now completed the anonymous round.

You invested 1 token(s) to the project and kept 19 token(s) for yourself.
You predicted that other members in your group will contribute 1 tokens.
You will earn payoffs from this anonymous round, but will only know the exact amount at the end of the experiment.

We will now proceed to the decision task.

(IV - Members) Stage 2: Introduction (Random, Delegate and Vote)

Round: 1
Period: 1

Your Information

Role: Member
ID: Dragonfruit

Group Information
Leader: [Unassigned]
Group members: Dragonfruit, Peach, Grape

Decision Task

You have been assigned a **Member** role.
Your unique ID is **Dragonfruit**

You have been randomly assigned to a group with 2 other members, Peach and Grape

You will also remain in this **same group** throughout all rounds in the decision task.

You and the 2 other participants will keep the **same group and ID** throughout all rounds in the decision task.

Next

Beginning of Round 1: Introduction Page 1/2

Round: 1
Period: 1

Your Information

Role: Member
ID: Dragonfruit

Group Information
Leader: [Unassigned]
Group members: Dragonfruit, Peach, Grape

Decision Task

We will now begin the decision task.

Each round will consist of 10 periods.

Before we begin Period 1 of Round 1, leaders will be assigned to groups of members via an assignment procedure.

Click proceed to continue.

Proceed

Beginning of Round 1: Introduction Page 2/2

Round: 2

Period: 1

Your Information



Role: Member

ID: Dragonfruit

Group Information

Leader: [Unassigned]

Group members:

Dragonfruit

Peach

Grape

Decision Task

We will now begin Round 2.

Note that you will remain in the same group, with Peach and Grape, throughout all rounds in the decision task.

A **new** pool of candidates will now be assigned to your group.

Like the previous round, before we begin Period 1, leaders will be assigned to groups of members via an assignment procedure.

Proceed

Beginning of Rounds 2-3: Introduction Page 1/1

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(V-1 - Members) Stage 2: Leader Assignment (Random)

Round: 1
Period: 1

Your Information

Role: Member
ID: Papaya

Group Information
Leader: [Unassigned]
Group members: Papaya, Mango, Durian

Leader assignment

Your group will now be assigned a leader from the new pool of candidates.
The leader will be assigned based on a random draw by the computer.

Next

Beginning of all rounds: Leader Assignment Page 1/3

Round: 1
Period: 1

Your Information

Role: Member
ID: Papaya

Group Information
Leader: [Unassigned]
Group members: Papaya, Mango, Durian

Leader assignment

The computer will randomly assign one of the following candidates as the leader of your group.
(Their IDs have been kept anonymous for now)

Anonymous Candidate	Score	Ranking
Candidate 1	2	Top 1/3 
Candidate 2	1	Middle 1/3 
Candidate 3	2	Top 1/3 

Hypothetically, if you were able to select the leader of your group, please state your order of preference of the above candidates.

1st: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
2nd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
3rd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3

Submit

Beginning of all rounds: Leader Assignment Page 2/3

Leader assignment result

Round: 1
Period: 1

Your Information



Role: Member
ID: Guava

Group Information

Leader: Yellow
Group members: Raspberry, Pear, Guava

In this round, leaders were assigned from the following pool of candidates based on a random draw by the computer.
Based on this procedure, Candidate 3: Yellow has been assigned as the leader of your group.

Candidate Pool's Addition Task Performance

Candidate	Score	Ranking
Black	2	Top 1/3 
Purple	1	Middle 1/3 
Yellow	2	Top 1/3 

Your leader will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

Beginning of all rounds: Leader Assignment Page 3/3

(V-2 - Members) Stage 2: Leader Assignment (Delegate)

Round: 1
Period: 1

Your Information

Role: Member
ID: Blueberry

Group Information
Leader: [Unassigned]
Group members: Blueberry, Lemon, Strawberry

Leader assignment

Your group will now be assigned a leader from the new pool of candidates.
The leader will be assigned based on the experimenter's preferences.

Next

Beginning of Round 1: Leader Assignment Page 1/3

Round: 1
Period: 1

Your Information

Role: Member
ID: Blueberry

Group Information
Leader: [Unassigned]
Group members: Blueberry, Lemon, Strawberry

Leader assignment

The experimenter will now assign one of the following candidates as the leader of your group.
(Their IDs have been kept anonymous for now)

Candidate Pool's Addition Task Performance		
Anonymous Candidate	Score	Ranking
Candidate 1	0	Middle 1/3 
Candidate 2	0	Middle 1/3 
Candidate 3	0	Middle 1/3 

Hypothetically, if you were able to select the leader of your group, please state your order of preference of the above candidates.

1st: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
2nd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
3rd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3

Submit

Beginning of Round 1: Leader Assignment Page 2/3

Round: 1
Period: 1

Your Information

Role: Leader
ID: Green

Group Information
Leader: Green
Group members: Coconut, Cherry, Banana

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on the experimenter's preferences.
Based on this procedure, you have been assigned as a leader for the group of Coconut, Cherry & Banana.

Candidate Pool's Addition Task Performance		
Candidate	Score	Ranking
Red	0	Middle 1/3 
Pink	0	Middle 1/3 
Green	0	Middle 1/3 

You will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

Beginning of Round 1: Leader Assignment Page 3/3

Round: 2
Period: 1

Your Information

Role: Member
ID: Blueberry

Group Information
Leader: [Unassigned]
Group members: Blueberry, Lemon, Strawberry

Leader assignment

Your group will now be assigned a leader from the new pool of candidates.
The leader will be assigned based on your past leader, Red's, preferences.

Next

Beginning of Rounds 2-3: Leader Assignment Page 1/3

Round: 2

Period: 1

Your Information



Role: Member

ID: Blueberry

Group Information

Leader: [Unassigned]

Group members:

Blueberry

Lemon

Strawberry

Leader assignment

Your past leader Red, will now assign one of the following candidates as the leader of your group.
(Their IDs have been kept anonymous for now)

Candidate Pool's Addition Task Performance		
Anonymous Candidate	Score	Ranking
Candidate 1	2	Top 1/3 
Candidate 2	1	Middle 1/3 
Candidate 3	2	Top 1/3 

Hypothetically, if you were able to select the leader of your group, please state your order of preference of the above candidates.

1st: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
2nd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
3rd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3

Submit

Beginning of Rounds 2-3: Leader Assignment Page 2/3

Round: 2

Period: 1

Your Information



Role: Leader

ID: Purple

Group Information

Leader: Purple

Group members:

Coconut

Cherry

Banana

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on the your group's past leader's preferences.

Based on this procedure, you have been assigned as a leader for the group of Coconut, Cherry & Banana.

Candidate Pool's Addition Task Performance		
Candidate	Score	Ranking
Black	2	Top 1/3 
Purple	1	Middle 1/3 
Yellow	2	Top 1/3 

You will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

Beginning of Rounds 2-3: Leader Assignment Page 3/3

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(V-3 - Members) Stage 2: Leader Assignment (Vote)

Round: 1
Period: 1

Your Information

Role: Member
ID: Apple

Group Information
Leader: [Unassigned]
Group members: Kiwi, Watermelon, Apple

Leader assignment

Your group will now be assigned a leader from the new pool of candidates.
The leader will be assigned based on your group's preferences.

Next

Beginning of all rounds: Leader Assignment Page 1/3

Round: 1
Period: 1

Your Information

Role: Member
ID: Apple

Group Information
Leader: [Unassigned]
Group members: Kiwi, Watermelon, Apple

Leader assignment

Members of your group will now decide on which of the following candidates they prefer as their leader.
(Their IDs have been kept anonymous for now)

Candidate Pool's Addition Task Performance		
Anonymous Candidate	Score	Ranking
Candidate 1	2	Top 1/3 
Candidate 2	0	Middle 1/3 
Candidate 3	0	Middle 1/3 

To select the leader of your group, please state your order of preference of the above candidates.

1st: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
2nd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3
3rd: ☐ Candidate 1 ☐ Candidate 2 ☐ Candidate 3

Submit

Beginning of all rounds: Leader Assignment Page 2/3

Round: 1
Period: 1

Your Information

Role: Member
ID: Dragonfruit

Group Information
Leader: White
Group members: Dragonfruit, Peach, Grape

Leader assignment result

In this round, leaders were assigned from the following pool of candidates based on your group's preferences. Based on this procedure, **Candidate 1: White** has been assigned as the leader of your group.

Candidate	Score	Ranking
White (Candidate 1)	0	Middle 1/3 
Blue (Candidate 2)	2	Top 1/3 
Brown (Candidate 3)	0	Middle 1/3 

Your leader will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

Beginning of all rounds: Leader Assignment Page 3/3

(VI - Members) Stage 2: Decision Task (Random, Delegate, Vote)

Round: 1
Period: 1

Your Information

Role: Member
ID: Blueberry

Group Information
Leader: Red
Group members: Blueberry, Lemon, Strawberry

Decision task

Messages can be sent by your leader in this period.
Your leader, Red, is currently deciding on the public message to send to you and your group members.

Type NEXT in the box below and click submit to proceed

Submit

All rounds, Odd Periods: Decision Task Page 1/2

Round: 1
Period: 1

Your Information

Role: Member
ID: Dragonfruit

Group Information
Leader: White
Group members: Dragonfruit, Peach, Grape

Decision task

Your leader, White, has sent you the following message .

It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute 1 tokens.
Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.

You have received an endowment of 20 tokens for this period.

Please enter the number of tokens you would like to invest in the joint project.

Your investment in the project:

Please also guess the average number of tokens the other members in your group will invest in this period.

Other members' average investment:

Submit

All rounds, Odd Periods: Decision Task Page 2/2

Round: 1
Period: 2

Your Information

Role: Member
ID: Blueberry

Group Information
Leader: Red
Group members: Blueberry, Lemon, Strawberry

Decision task

Messages cannot be sent by your leader in this period.

Your leader, Red, has sent you the following message [in the previous period](#)

It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute 1 tokens.

Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.

You have received an endowment of 20 tokens for this period.

Please enter the number of tokens you would like to invest in the joint project.

Your investment in the project:

Please also guess the average number of tokens the other members in your group will invest in this period.

Other members' average investment:

All rounds, Even Periods: Decision Task Page

Round: 1
Period: 1

Your Information

Role: Member
ID: Blueberry

Group Information
Leader: Red
Group members: Blueberry, Lemon, Strawberry

Decision task - summary

Below is a summary of the investments all members have made during this period.

Member	Investment
Blueberry (you)	1
Lemon	1
Strawberry	1

You invested 1 token(s) to the project and kept 19 token(s) for yourself.
A total of 3 token(s) were contributed to the project.
Your earnings from the decision task is 21.0 token(s).

Your prediction on the average contribution of others is 1.0 token(s).
The actual average contribution of others is 1.0 token(s).
The reward from your prediction is 0.5 token(s).

All rounds, All Periods: Decision Task Results

End of Round 1

You have now completed Round 1.

Before we begin the next round, please answer the following questions.

1. To what extent do you think your leader was important in improving your group's investments in this round?

☐ Not at all ☐ Slightly ☐ Moderately ☐ Very ☐ Extremely

2. What is your opinion about your leader in this round?

☐ Very Negative ☐ Negative ☐ Neutral ☐ Positive ☐ Very Positive

3. To what extent do you think the public message mattered in your decision?

☐ Not at all ☐ Slightly ☐ Moderately ☐ Very ☐ Extremely

Submit

All Rounds, Post-Round Questions

(IV - Candidates) Stage 2: Introduction (Random, Delegate, Vote)

Round: 1
Period: 1

Your Information

Role: Candidate
ID: Pink

Group Information
Leader: [Unassigned]
Group members: [Unassigned]

Decision Task

You have been assigned a **Candidate** role.
Your unique ID is **Pink**.

You have been randomly assigned to a candidate pool with 2 other candidates, Green and Red.

You will remain in the **same pool of candidates** throughout all rounds in the decision task.

You and the other 2 participants will keep the **same role and ID** throughout all rounds in the decision task.

Next

Beginning of Round 1: Candidate Page 1/2

Round: 1
Period: 1

Your Information

Role: Candidate
ID: Pink

Group Information
Leader: [Unassigned]
Group members: [Unassigned]

Decision Task

We will now begin the decision task.

Each round will consist of 10 periods.

Before we begin Period 1 of Round 1, leaders will be assigned to groups of members via an assignment procedure.

Click proceed to continue.

Proceed

Beginning of Round 1: Candidate Page 2/2

Round: 2

Period: 1

Your Information

Role: Candidate

ID: Pink

Group Information

Leader: [Unassigned]

Group members: [Unassigned]

Decision Task

We will now begin Round 2.

Note that you will remain in the same pool of candidates, with Green and Red, throughout all rounds in the decision task.

Your pool of candidates will now be assigned to a **new** group.

Like the previous round, before we begin Period 1, leaders will be assigned to groups of members via an assignment procedure.

Proceed

Beginning of Rounds 2-3: Candidate Page

(V-1 - Candidates) Stage 2: Leader Assignment Result (Random)

Round: 1
Period: 1

Your Information

Role: Leader
ID: Black

Group Information
Leader: Black
Group members: Papaya, Mango, Durian

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on a random draw by the computer.
Based on this procedure, you have been assigned as a leader for the group of Papaya, Mango & Durian.

Candidate	Score	Ranking
Black	2	Top 1/3 
Purple	1	Middle 1/3 
Yellow	2	Top 1/3 

You will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

All Rounds: Leader Assignment Result (Assigned)

Round: 1
Period: 1

Your Information

Role: Shadow leader
ID: Purple

Group Information
Leader: Black
Group members: Papaya, Mango, Durian

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on a random draw by the computer.
Based on this procedure, you have not been assigned as a leader to any group.

Candidate	Score	Ranking
Black	2	Top 1/3 
Purple	1	Middle 1/3 
Yellow	2	Top 1/3 

You will be asked to make **hypothetical decisions** of a public message to send every odd-numbered periods.

Proceed

All Rounds: Leader Assignment Result (Not Assigned)

(V-2 - Candidates) Stage 2: Leader Assignment Result (Delegate)

Round: 1
Period: 1

Your Information

Role: Leader
ID: Green

Group Information
Leader: Green
Group members: Coconut, Cherry, Banana

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on the experimenter's preferences.
Based on this procedure, you have been assigned as a leader for the group of Coconut, Cherry & Banana.

Candidate	Score	Ranking
Red	0	Middle 1/3 
Pink	0	Middle 1/3 
Green	0	Middle 1/3 

You will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

Round 1: Leader Assignment Result (Assigned)

Round: 1
Period: 1

Your Information

Role: Shadow leader
ID: Pink

Group Information
Leader: Red
Group members: Blueberry, Lemon, Strawberry

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on the experimenter's preferences.
Based on this procedure, you have not been assigned as a leader to any group.

Candidate	Score	Ranking
Red	0	Middle 1/3 
Pink	0	Middle 1/3 
Green	0	Middle 1/3 

You will be asked to make **hypothetical decisions** of a public message to send every odd-numbered periods.

Proceed

Round 1: Leader Assignment Result (Not Assigned)

Round: 2
Period: 1

Your Information

Role: Leader
ID: Purple

Group Information
Leader: Purple
Group members: Coconut, Cherry, Banana

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on the your group's past leader's preferences.
Based on this procedure, you have been assigned as a leader for the group of Coconut, Cherry & Banana.

Candidate	Score	Ranking
Black	2	Top 1/3 
Purple	1	Middle 1/3 
Yellow	2	Top 1/3 

You will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

Rounds 2-3: Leader Assignment Result (Assigned)

Round: 2
Period: 1

Your Information

Role: Shadow leader
ID: Black

Group Information
Leader: Purple
Group members: Coconut, Cherry, Banana

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on the group's past leader's preferences.
Based on this procedure, you have not been assigned as a leader to any group.

Candidate	Score	Ranking
Black	2	Top 1/3 
Purple	1	Middle 1/3 
Yellow	2	Top 1/3 

You will be asked to make **hypothetical decisions** of a public message to send every odd-numbered periods.

Proceed

Rounds 2-3: Leader Assignment Result (Not Assigned)

(V-3 - Candidates) Stage 2: Leader Assignment Result (Vote)

Round: 1
Period: 1

Your Information

Role: Leader
ID: Blue

Group Information
Leader: Blue
Group members: Kiwi, Watermelon, Apple

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on group members' preferences.
Based on this procedure, you have been assigned as a leader for the group of Kiwi, Watermelon & Apple.

Candidate	Score	Ranking
White	0	Middle 1/3 
Brown	0	Middle 1/3 
Blue	2	Top 1/3 

You will be asked to make decisions of a public message to send every odd-numbered periods.

Proceed

All Rounds: Leader Assignment Result (Assigned)

Round: 1
Period: 1

Your Information

Role: Shadow leader
ID: Brown

Group Information
Leader: White
Group members: Dragonfruit, Peach, Grape

Leader assignment result

In this round, leaders were assigned from your pool of candidates based on group members' preferences.
Based on this procedure, you have not been assigned as a leader to any group.

Candidate	Score	Ranking
White	0	Middle 1/3 
Brown	0	Middle 1/3 
Blue	2	Top 1/3 

You will be asked to make **hypothetical decisions** of a public message to send every odd-numbered periods.

Proceed

All Rounds: Leader Assignment Result (Not Assigned)

(VI - Assigned Leaders) Stage 2: Decision Task (Random, Delegate, Vote)

Round: 1
Period: 1

Your Information

Role: Leader
ID: White

Group Information
Leader: White
Group members: Dragonfruit, Peach, Grape

Decision task

Please decide on the message you want to send to your members.
Do also key in a whole number for the recommended number of tokens to contribute.

Message Preview

It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute the following amount of tokens.

Please select one of the following supplementary messages.

- ☐ Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.
- ☐ We need to reverse this downward trend. By following my strategy, we can achieve a better outcome if we all invest more.
- ☐ We are getting there, but if you follow my guidelines and invest more, we will all benefit.
- ☐ We have achieved my goal of high investments and high earnings, which we must sustain.
- ☐ This situation is not following my aims for the group. We should change course and invest more.
- ☐ You should be ashamed of not following my instructions.
- ☐ You have done well to follow my instructions.
- ☐ We are trying to maximise our investments and earnings. Unless you follow my instructions, you are not part of the group.
- ☐ (No message)

Submit

All Rounds, Odd Periods: Decision Task Page 1/2 [Personal Legitimacy]

Round: 1
Period: 1

Your Information

Role: Leader
ID: White

Group Information
Leader: White
Group members: Dragonfruit, Peach, Grape

Decision task

You have sent the following message .

It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute 1 tokens.
Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.

The members receive an endowment of 20 tokens for this period.
They are currently deciding on their investments to the joint project.

Please guess the average number of tokens the members in your group will invest in this period.

Members' average investments:

Submit

All Rounds, Odd Periods: Decision Task Page 2/2 [Personal Legitimacy]

Round: 1
Period: 1

Your Information



Role: Leader
ID: Yellow

Group Information

Leader: Yellow

Group members: Papaya, Mango, Durian

Decision task

Please decide on the message you want to send to your members.
Do also key in a whole number for the recommended number of tokens to contribute.

Message Preview

It is my role to tell you how you can draw from norms in society in order to achieve high levels of investments and earnings. Speaking as a similar member of society, I suggest that each of you contribute the following amount of tokens:

19

This situation is not following my aims for the group. We should change course and invest more.

Please select one of the following supplementary messages.

- ☐ Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that society has laid out for us.
- ☐ We need to reverse this downward trend. By following norms in society, we can achieve a better outcome if we all invest more.
- ☐ We are getting there, but if we think more about our societal standards and invest more, we will all benefit.
- ☐ We have achieved a goal worthy of members of society which we must sustain.
- ☒ This situation is not befitting the norms of society. We should change course and invest more.
- ☐ You should be ashamed of your actions which violate norms of society.
- ☐ You have done well to uphold the norms of society.
- ☐ We are trying to maximise our investments and earnings. Unless you abide by the norms of society, you are not part of the group.
- ☐ (No message)

Submit

All Rounds, Odd Periods: Decision Task Page 1/2 [Institutional Legitimacy]

Round: 1
Period: 1

Your Information



Role: Leader
ID: White

Group Information

Leader: White

Group members: Kiwi, Watermelon, Apple

Decision task

You have sent the following message .

It is my role to tell you how you can draw from norms in society in order to achieve high levels of investments and earnings. Speaking as a similar member of society, I suggest that each of you contribute 20 tokens. We need to reverse this downward trend. By following my strategy, we can achieve a better outcome if we all invest more.

The members receive an endowment of 20 tokens for this period.
They are currently deciding on their investments to the joint project.

Please guess the average number of tokens the members in your group will invest in this period.

Members' average investments:

Submit

All Rounds, Odd Periods: Decision Task Page 2/2 [Institutional Legitimacy]

Round: 1
Period: 2

Your Information

Role: Leader
ID: Yellow

Group Information
Leader: Yellow
Group members: Raspberry, Pear, Guava

Decision task

Messages cannot be sent to the members in this period.

You have sent the following message [in the previous period](#).

It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute 1 tokens. Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.

The members receive an endowment of 20 tokens for this period. They are currently deciding on their investments to the joint project.

Please guess the average number of tokens the members in your group will invest in this period.

Members' average investments:

All Rounds, Even Periods: Decision Task Page [Personal Legitimacy]

Round: 1
Period: 2

Your Information

Role: Member
ID: Papaya

Group Information
Leader: Yellow
Group members: Papaya, Mango, Durian

Decision task

Messages cannot be sent by your leader in this period.

Your leader, Yellow, has sent you the following message [in the previous period](#).

It is my role to tell you how you can draw from norms in society in order to achieve high levels of investments and earnings. Speaking as a similar member of society, I suggest that each of you contribute 19 tokens. This situation is not following my aims for the group. We should change course and invest more.

You have received an endowment of 20 tokens for this period.

Please enter the number of tokens you would like to invest in the joint project.

Your investment in the project:

Please also guess the average number of tokens the other members in your group will invest in this period.

Other members' average investment:

All Rounds, Even Periods: Decision Task Page [Institutional Legitimacy]

Round: 1

Period: 1

Your Information



Role: Leader

ID: Black

Group Information

Leader: Black

Group members:

Papaya

Mango

Durian

Decision task - summary

Below is a summary of the investments all members have made during this period.

Member	Investment
Papaya	1
Mango	1
Durian	1

A total of 3 token(s) were contributed to the project.

Your earnings for the decision task is 16.1 token(s).

Your prediction on the average contribution is 1.0 token(s).

The actual average contribution is 1.0 token(s).

The reward from your prediction is 0.5 token(s).

Proceed

All Rounds, All Periods: Decision Task Result

End of Round 1

You have now completed Round 1.

Before we begin the next round, please answer the following questions.

1.To what extent do you think your followers listened to your instructions?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

2.To what extent was your message effective?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

3.To what extent do you think you were a good leader?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

4.To what extent do you think your followers think of you as a good leader?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

Submit

All Rounds, Post-Round Questions

(VI - Shadow Leaders) Stage 2: Decision Task (Random, Delegate, Vote)

Round: 1
Period: 1

Your Information

Role: Shadow leader
ID: Pink

Group Information
Leader: Red
Group members: Blueberry, Lemon, Strawberry

Decision task

You are to make a hypothetical decision on the public message you would like to send to your members.
Please decide on the message you want to send to your members.
Do also key in a whole number for the recommended number of tokens to contribute.

Message Preview
It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute the following amount of tokens:

Please select one of the following supplementary messages.

- ☐ Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.
- ☐ We need to reverse this downward trend. By following my strategy, we can achieve a better outcome if we all invest more.
- ☐ We are getting there, but if you follow my guidelines and invest more, we will all benefit.
- ☐ We have achieved my goal of high investments and high earnings, which we must sustain.
- ☐ This situation is not following my aims for the group. We should change course and invest more.
- ☐ You should be ashamed of not following my instructions.
- ☐ You have done well to follow my instructions.
- ☐ We are trying to maximise our investments and earnings. Unless you follow my instructions, you are not part of the group.
- ☐ (No message)

Submit

All Rounds, Odd Periods: Decision Task Page 1/2 [Personal Legitimacy]

Round: 1
Period: 1

Your Information

Role: Shadow leader
ID: Pink

Group Information
Leader: Red
Group members: Blueberry, Lemon, Strawberry

Decision task

The leader of the group, Red, sent the following message.

It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute 1 tokens.
Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.

The members receive an endowment of 20 tokens for this period.
They are currently deciding on their investments to the joint project.

Please guess the average number of tokens the members in your group will invest in this period.

Members' average investments:

Submit

All Rounds, Odd Periods: Decision Task Page 2/2 [Personal Legitimacy]

Round: 1
Period: 1

Your Information

Role: Shadow leader
ID: Blue

Group Information
Leader: White
Group members: Kiwi, Watermelon, Apple

Decision task

You are to make a hypothetical decision on the public message you would like to send to your members. Please decide on the message you want to send to your members. Do also key in a whole number for the recommended number of tokens to contribute.

Message Preview
It is my role to tell you how you can draw from norms in society in order to achieve high levels of investments and earnings. Speaking as a similar member of society, I suggest that each of you contribute the following amount of tokens:

20

We have achieved my goal of high investments and high earnings, which we must sustain.

Please select one of the following supplementary messages.

- ☐ Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that society has laid out for us.
- ☐ We need to reverse this downward trend. By following norms in society, we can achieve a better outcome if we all invest more.
- ☐ We are getting there, but if we think more about our societal standards and invest more, we will all benefit.
- ☒ We have achieved a goal worthy of members of society which we must sustain.
- ☐ This situation is not befitting the norms of society. We should change course and invest more.
- ☐ You should be ashamed of your actions which violate norms of society.
- ☐ You have done well to uphold the norms of society.
- ☐ We are trying to maximise our investments and earnings. Unless you abide by the norms of society, you are not part of the group.
- ☐ (No message)

Submit

All Rounds, Odd Periods: Decision Task Page 1/2 [Institutional Legitimacy]

Round: 1
Period: 1

Your Information

Role: Shadow leader
ID: Blue

Group Information
Leader: White
Group members: Kiwi, Watermelon, Apple

Decision task

The leader of the group, White, sent the following message .

It is my role to tell you how you can draw from norms in society in order to achieve high levels of investments and earnings. Speaking as a similar member of society, I suggest that each of you contribute 20 tokens. We need to reverse this downward trend. By following my strategy, we can achieve a better outcome if we all invest more.

The members receive an endowment of 20 tokens for this period. They are currently deciding on their investments to the joint project.

Please guess the average number of tokens the members in your group will invest in this period.

Members' average investments:

Submit

All Rounds, Odd Periods: Decision Task Page 2/2 [Institutional Legitimacy]

Round: 1

Period: 2

Your Information



Role: Shadow leader

ID: Purple

Group Information

Leader: Black

Group members:

Papaya

Mango

Durian

Decision task

Messages cannot be sent in this period.

The leader of the group, Black, sent the following message [in the previous period](#).

It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings. Speaking as leader of your group, I suggest that each of you contribute 1 tokens.

Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.

The members receive an endowment of 20 tokens for this period.

They are currently deciding on their investments to the joint project.

Please guess the average number of tokens the members in your group will invest in this period.

Members' average investments:

Submit

All Rounds, Even Periods: Decision Task Page [Personal Legitimacy]

Round: 1

Period: 2

Your Information



Role: Shadow leader

ID: Black

Group Information

Leader: Yellow

Group members:

Papaya

Mango

Durian

Decision task

Messages cannot be sent in this period.

The leader of the group, Yellow, sent the following message [in the previous period](#).

It is my role to tell you how you can draw from norms in society in order to achieve high levels of investments and earnings. Speaking as a similar member of society, I suggest that each of you contribute 19 tokens.

This situation is not following my aims for the group. We should change course and invest more.

The members receive an endowment of 20 tokens for this period.

They are currently deciding on their investments to the joint project.

Please guess the average number of tokens the members in your group will invest in this period.

Members' average investments:

Submit

All Rounds, Even Periods: Decision Task Page [Institutional Legitimacy]

55

Round: 1

Period: 1

Your Information



Role: Shadow leader

ID: Purple

Group Information

Leader: Black

Group members:

Papaya

Mango

Durian

Decision task - summary

Below is a summary of the investments all members have made during this period.

Member	Investment
Papaya	1
Mango	1
Durian	1

A total of 3 token(s) were contributed to the project.

Your earnings for the decision task is 11.1 token(s).

Your prediction on the average contribution is 1.0 token(s).

The actual average contribution is 1.0 token(s).

The reward from your prediction is 0.5 token(s).

Proceed

All Rounds, All Periods: Decision Task Result

End of Round 1

You have now completed Round 1.

Before we begin the next round, please answer the following questions.

1.To what extent do you think members listened to Black's instructions?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

2.To what extent was Black's message effective?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

3.To what extent do you think Black was a good leader?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

4.To what extent do you think the followers think that Black is a good leader?

☐ Not at all
☐ Slightly
☐ Moderately
☐ Very
☐ Extremely

Submit

All Rounds, Post-Round Questions

(VII) Payment Summary (Random, Delegate, Vote)

Earnings Summary

Anonymous Round

Earnings: 36.0 tokens

Addition Task

Earnings: 0.0 tokens

Decision Task

ROUND 1			ROUND 2			ROUND 3		
Period	Tokens	Selected	Period	Tokens	Selected	Period	Tokens	Selected
1	16.1		1	16.1	X	1	16.1	
2	16.1		2	16.1		2	16.1	
3	16.1		3	16.1		3	16.1	
4	16.1		4	16.1		4	16.1	
5	16.1		5	16.1	X	5	16.1	X
6	16.1		6	16.1		6	16.1	
7	16.1		7	16.1		7	16.1	X
8	16.1	X	8	16.1	X	8	16.1	
9	16.1	X	9	16.1		9	16.1	X
10	16.1	X	10	16.1		10	16.1	

Total tokens earned (Anonymous round + Addition task + Decision task):

= 36.0 + 0.0 + 144.9

= 180.9 = SGD\$ 9.04

Total payment (SGD) (Show up fee + Earnings from all tasks):

= \$2.0 + \$9.04

= \$ 11.5

Proceed

End of Experiment (after Post-Experiment Questionnaire): Earnings

57

D.3 Post-Round Survey

For Leaders

To what extent do you think your followers listened to your instructions? [1: *Not at all* - 5: *Extremely*]

To what extent was your message effective? [1: *Not at all* - 5: *Extremely*]

To what extent do you think you were a good leader? [1: *Not at all* - 5: *Extremely*]

To what extent do you think your followers think of you as a good leader? [1: *Not at all* - 5: *Extremely*]

For Unassigned Candidates

To what extent do you think members listened to [assigned leader's name]'s instructions?

[1: *Not at all* - 5: *Extremely*]

To what extent was [assigned leader's name]'s message effective? [1: *Not at all* - 5: *Extremely*]

To what extent do you think [assigned leader's name] was a good leader? [1: *Not at all* - 5: *Extremely*]

To what extent do you think the followers think that [assigned leader's name] is a good leader?

[1: *Not at all* - 5: *Extremely*]

For Members

To what extent do you think your leader was important in improving your group's investments in this round?

[1: *Not at all* - 5: *Extremely*]

What is your opinion about your leader in this round? [1: *Very Negative* - 5: *Very Positive*]

To what extent do you think the public message mattered in your decision? [1: *Not at all* - 5: *Extremely*]

D.4 Post-Experiment Survey

Demographic questions

Gender [1: Male; 2: Female; 3: I would prefer not to say]

Course of study [Open-ended]

Year of study [1: Year 1; 2: Year 2; 3: Year 3; 4: Year 4; 5: Year 5; 6: Postgraduate; 7: Not Applicable]

Nationality [Open-ended]

Ethnicity [Open-ended]

Cultural values

Please indicate the extent to which you agree or disagree with the following statements.

[1: Strongly disagree - 4: Strongly agree]

Authority (Feldman, 2003)

Children should be taught to obey authority in schools.

Leaders know what is best for followers/others.

The most important values that should be taught to children are obedience to authority and respect of authority.

Collectivism (Torelli and Shavitt, 2010; Shavitt et al., 2006)

Cooperating with others makes me feel good.

It is my duty to take care of my family, even if it involves self-sacrifice of what I want.

When I am a member in groups, I appreciate what the majority wants and wishes to do.

Key factors for getting promotions are hardwork and persistence, rather than who you are.

Locus of Control (Rotter, 1966)

Leadership is not a talent; most people can learn to become leaders.

I am very persevering and usually accomplish what I set out to do.

There is little point in making plans when the future can't be predicted.

Competitiveness (Triandis and Gelfand, 1998)

To me, winning is everything.

I feel tensed and aroused when someone else does something better than me.

Working on my own makes it more likely of me to accomplish my goals.

Demand for punishment (Altemeyer, 1988; Duckitt and Sibley, 2010)

Young people today don't respect traditional values enough.

People no longer know how to differentiate between right and wrong.

Stiffer sentences should be given to people who break the law.

Legitimacy source

An assigned leader should obey the implicit rules agreed on by the group.

An assigned leader has the right to set the rules which govern the group.

A leader only has power over his followers to the extent that they support him.

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General questions

Was there anything unclear about the instructions? If so, please state so. [Open-ended]

Did you find anything unfair about the experiment? If so, please state so. [Open-ended]

Questions for Candidates

As a candidate, what do you think was the most important trait to be assigned as a leader? [Open-ended]

If you were a leader at some point, how did you decide on your recommendation to members? [Open-ended]

Questions for Members

As a member, what determined which candidate you preferred? [Open-ended]

As a member, how did you decide how much to contribute to the joint project at any one point? [Open-ended]

How did you decide whether to follow the leader's instruction to contribute to the joint project? [Open-ended]

Preference for leadership style

Suppose you were a leader and given a choice between two sets of messages to send to your members: Set A and Set B. Which set would you prefer?

Set A

1. Speaking as a similar member of society, I suggest that each of you contributes X.
2. It is my role to tell you how you can draw from norms in society in order to achieve high levels of investments and earnings.
3. Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that society has laid out for us.
4. We need to reverse this downward trend. By following norms in society, we can achieve a better outcome if we all invest more.
5. We are getting there, but if we think more about our societal standards and invest more, we will all benefit.
6. We have achieved a goal worthy of members of society which we must sustain.
7. This situation is not befitting the norms of society. We should change course and invest more.
8. You should be ashamed of your actions which violate norms of society.
9. You have done well to uphold the norms of society.
10. We are trying to maximise our investments and earnings. Unless you abide by the norms of society, you are not part of the group.

Set B

1. Speaking as leader of your group, I suggest that each of you contributes X.
 2. It is my role to tell you how you can draw from my experience in order to achieve high levels of investments and earnings.
 3. Let's keep it up. We have been successful in the last round, and you should continue to follow this high investment path that I have laid out.
 4. We need to reverse this downward trend. By following my strategy, we can achieve a better outcome if we all invest more.
 5. We are getting there, but if you follow my guidelines and invest more, we will all benefit.
 6. We have achieved my goal of high investments and high earnings, which we must sustain.
 7. This situation is not following my aims for the group. We should change course and invest more.
 8. You should be ashamed of not following my instructions.
 9. You have done well to follow my instructions.
 10. We are trying to maximise our investments and earnings. Unless you follow my instructions, you are not part of the group.
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